MaxwellFD: 2D Yee FDTD and FDFD

Physics, methods, and code

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Abstract

MaxwellFD is a Python Yee-cell solver. One uniform Cartesian grid, one material sampling, and one pair of discrete curls feed both a leapfrog (FDTD) and a sparse frequency-domain system (FDFD). Phasors use $e^{j\omega t}$, the same convention as MaxwellMOM. This note describes the physics and the method that are actually coded, how the package is organized, and how the Fresnel-slab, dielectric-cylinder, PML line-current, and open PEC-cylinder examples are run and drawn. The equation reference is `FD_LaTeX_Reference.tex`. The convolutional PML from that note is coded as a stretch of the normal derivatives on a PEC-backed band. Embedded PEC and PMC objects remove Yee samples, and an open PEC cylinder is compared with the Mie series on that band. An open dielectric cylinder on the same band is driven by the contrast current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$. A 3D PEC cavity checks three pure modes of the Yee cell. An open dielectric sphere, on a PML small enough to factor, is compared with the Mie series. A resistive sheet adds its conductance to one Ampere cell and is compared with the analytic reflection. A symmetric GSTC sheet replaces the magnetic sample on one face and is compared the same way. A finite run of that face, with $\chi_{ee} = \chi_{mm} = -2j/k$, is a normal-incidence absorber on a four-sided PML. The normal magnetic susceptibility χ^{nn}_{mm} enters the TMz jump, and one periodic cell at an angle is compared with the analytic reflection and transmission. The same susceptibilities on a 10λ sheet at $30°$ are the open scattered-field run. A piecewise sheet gives each magnetic sample the susceptibilities of the piece that contains its x. Polynomial electric dispersion $\chi_{ee0} + \chi_{ee2}k_t^2$ is the normal susceptibility $\chi^{nn}_{mm} = \chi_{ee2}k^2$. Polynomial magnetic dispersion $\chi_{mm0} + \chi_{mm2}k_t^2$ is a second difference of the appended tangential H, in both polarizations. The denominator $1 + \xi_{mm2}k_t^2$ is an auxiliary unknown on that jump.

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1 Status

Version 0.1.0 contains the following.

1. The 2D cell in free space and in a uniform isotropic medium: TMz and TEz curls, PEC and periodic walls, the CFL limit, cavity eigenvalues, a manufactured FDFD solve, and a periodic standing wave.

2. Volumetric media. A staircase assigns a real ε, μ, and $\sigma \geq 0$ to each Yee sample. FDFD accepts a complex $\varepsilon(\omega)$. FDTD adds one bulk Debye pole or one bulk Lorentz pole through a polarization current. The closed-box checks are a Fresnel slab and a dielectric cylinder, both driven by an analytic Dirichlet trace on the existing PEC wall.

3. The convolutional PML on a PEC-backed band, for both polarizations and both drivers. In FDFD every derivative carries the factor $1/s_w(\omega)$. The leapfrog keeps one auxiliary for each of those derivatives. A line current is compared with the outgoing Hankel function on the unstretched samples.

4. Embedded PEC and PMC objects. PEC drops every electric sample inside its region. PMC drops every magnetic sample. The open check is a PEC cylinder: the unknown is the scattered field, the object carries the trace $-E_{\text{inc}}$, and the PML absorbs the outgoing wave.

5. An open dielectric cylinder on that PML. The staircase disk stays in the operator. The incident wave is the impressed current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$, and a free sample of the total field is $E_s + E_{\text{inc}}$.

6. The 3D Yee cell and a sparse FDFD operator on a rectangular box. Uniform isotropic media, with PEC and periodic walls. Three pure cavity modes are checked by a Rayleigh quotient against the semi-discrete frequency with the third centered difference.

7. An open dielectric sphere. The 3D FDFD operator stretches a PEC-backed PML and keeps a staircase ball in the permittivity. The incident wave is the contrast current, the unknown is the scattered field, and the total electric field is compared with the Mie series.

8. A z-directed Hertzian dipole. One E_z sample carries $J_z = I\ell/(\Delta x \Delta y \Delta z)$. The electric field outside the PML, and at least $\lambda/4$ from the element, is compared with the outgoing spherical wave.

9. A 2D paraxial Gaussian beam. The impressed current on the two E_z rows beside a horizontal split launches the beam into the total-field side. That electric field, outside the PML and at least $\lambda/2$ from the split, is compared with the beam. The scattered-field side stays quiet.

10. A 2D TMz near-to-far integral. A closed contour in the unstretched interior carries E_z on the nodes and the averaged tangential H. The far field of a line current is compared with the leading Hankel wave.

11. A Bloch phase on the periodic x wrap. Stored samples are the physical field on $[0, a)$. The forward image carries $e^{-jk_{B,x}a}$ and the backward image carries $e^{+jk_{B,x}a}$. The factor is one when $k_{B,x} = 0$. A cell 0.8λ wide at 30° uses the susceptibilities of the phase-one oblique sheet and the analytic trace on the y ends. The packed electric field is compared with that trace.

12. A resistive sheet of normal $\hat{\mathbf{y}}$. The conductance $1/(Z_s \Delta y)$ is added to σ on the tangential electric samples of the one Ampere cell that contains the sheet. The closed-box check compares the packed electric field with the analytic reflection and transmission.

13. A symmetric GSTC sheet of normal $\hat{\mathbf{y}}$. The magnetic sample on one face is replaced by the pair H^- and H^+. Tangential χ_{ee} and χ_{mm} close the pair. The closed-box check compares the packed electric field with the analytic reflection and transmission. A half-open x interval keeps a finite run of the face. The jump stays unstretched, so the sheet may sit in a PML box and must stay out of the band. $\chi_{ee} = \chi_{mm} = -2j/k$ is a perfect absorber at normal incidence. The scattered-field check is a 4λ sheet. χ^{nn}_{mm} adds $\partial_x H_y$ to the TMz jump. H_y is not split. A periodic cell one transverse period wide, with the analytic trace on the y ends, checks that term at 30°. The same susceptibilities on a 10λ sheet at 30° are the open run of that face.

14. A piecewise GSTC sheet. Each magnetic sample takes the five susceptibilities of the piece that contains its x. The jumps stay the same. A 6λ sheet split into $3\lambda + 3\lambda$ compares each piece with that piece's analytic field, one wavelength inside the ends and the junction. The left piece is the normal-incidence absorber and the right piece is $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$. The 10λ run of that sheet is the open example.

15. Polynomial electric dispersion on a periodic cell. The susceptibility $\chi_{ee0} + \chi_{ee2} k_t^2$ is the normal term $\chi^{nn}_{mm} = \chi_{ee2} k^2$. TMz is compared with that susceptibility at $k \sin\theta$. TEz keeps χ_{ee0}.

16. Polynomial magnetic dispersion on a periodic cell. The susceptibility $\chi_{mm0} + \chi_{mm2} k_t^2$ is a second difference of the appended tangential H. Both polarizations are compared with that susceptibility at $k \sin\theta$.

17. A rational magnetic susceptibility on a periodic cell. The quotient $(\chi_{mm0} + \chi_{mm2} k_t^2)/(1 + \xi_{mm2} k_t^2)$ is one auxiliary per cut. Both polarizations are compared with that susceptibility at $k \sin\theta$.

MaxwellFD does not import MaxwellMOM. The shared pieces are the time convention, the four SI constants, and the analytic targets used as external checks. The 3D FDFD operator stretches a PEC-backed PML and accepts a staircase ball. The open checks are a dielectric sphere, a Hertzian dipole, a Gaussian beam, and the far field of a line current. A 3D leapfrog remains a later slice. Embedded PEC and PMC objects remove Yee samples on the 2D grid. The open PEC cylinder and the open dielectric cylinder are scattered-field solves on the PML.

2 Physics

Fields are in SI units. With the convention $e^{j\omega t}$, a wave $e^{-jk\hat{\mathbf{k}}\cdot\mathbf{r}}$ travels in the direction $\hat{\mathbf{k}}$, and ∂_t becomes $j\omega$. The constants in `maxwell_fd.utils.constants` are

$$c_0 = 299792458\,\mathrm{m\,s^{-1}}, \qquad \varepsilon_0 = 8.8541878128 \times 10^{-12}\,\mathrm{F\,m^{-1}},$$
$$\mu_0 = 1.25663706212 \times 10^{-6}\,\mathrm{H\,m^{-1}}, \qquad \eta_0 = 376.73031346177\,\Omega.$$

Faraday's law and Ampere's law with an impressed current $\mathbf{J}$ are

$$\nabla \times \mathbf{E} = -j\omega\mu\,\mathbf{H}, \tag{1}$$
$$\nabla \times \mathbf{H} = j\omega\varepsilon\,\mathbf{E} + \mathbf{J}. \tag{2}$$

A real conductivity is not a second impressed current. In FDFD it enters as $\varepsilon + \sigma/(j\omega)$ when ε is real. A law that already returns a complex $\varepsilon(\omega)$ has included its loss, and the accompanying σ must be zero.

The grid lies in the xy-plane and nothing depends on z. TMz keeps E_z, H_x, and H_y. TEz keeps H_z, E_x, and E_y. The code never forms the scalar Helmholtz operator. Both drivers eliminate, or update, the magnetic field through the curls, so a face-by-face μ stays in the magnetic samples.

On a PEC wall the tangential electric field is the boundary value stored with the grid. Milestone 1 sets that value to zero. The slab and the cylinder instead prescribe the analytic tangential trace and leave the impressed current at zero. A periodic wall identifies opposite faces. The forward x wrap multiplies that image by $e^{-jk_{B,x}a}$, and the backward wrap multiplies by $e^{+jk_{B,x}a}$. The factor is one when $k_{B,x} = 0$, which is the milestone-1 wall and the cell whose width is one transverse period. `YeeGrid2D` accepts `Boundary.PEC`, `Boundary.PERIODIC`, and `Boundary.PERIODIC_X`. The last of these is periodic in x and PEC in y. A nonzero $k_{B,x}$ needs one of those two periodic walls. The y identification stays at phase one. The leapfrog and the real eigenproblem refuse a nonzero Bloch wave number. A PML is the stretch of Sec. 3.5 on a band inside that PEC wall. A resistive sheet of normal $\hat{\mathbf{y}}$ adds $\sigma_{\text{eff}} = \sigma + 1/(Z_s \Delta y)$ to the tangential electric samples of the one Ampere cell that contains the sheet. That cell is the half-open interval $y - \Delta y/2 \le y_{\text{sheet}} < y + \Delta y/2$. Figure 1 draws that cell across the two primary cells that meet on the sheet. TMz loads E_z and TEz loads E_x. The normal electric sample is left alone. The leapfrog absorbs the added conductivity in the coefficients (11). FDFD absorbs it on the diagonal $\varepsilon + \sigma/(j\omega)$. A symmetric GSTC sheet of the same normal sits on a magnetic face, halfway between electric rows. Figure 2 draws that face in the primary cell the sheet cuts. The magnetic sample on the face is replaced by the pair H^- and H^+. Tangential χ_{ee} and χ_{mm} close the pair. FDFD appends the pair after the electric unknowns and returns the electric field. An optional half-open interval $x_0 \le x < x_1$ keeps a finite run of those samples. The default is the whole face. The jump uses the unstretched stencil, so a cut inside the PML is refused. χ^{nn}_{mm} adds $\chi^{nn}_{mm}\partial_x H_{y,\text{av}}$ on the TMz side of that jump and leaves TEz unchanged. On a periodic x face the wrapped neighbors of that derivative, and the wrapped TEz link at the right edge, carry the same Bloch factor. A piecewise sheet assigns each magnetic sample the susceptibilities of the piece that contains its x, including χ_{mm2} and ξ_{mm2} when that piece sets them. The jumps stay the same, and a sample outside every piece stays in the bulk curl. Polynomial electric dispersion $\chi_{ee0} + \chi_{ee2}k_t^2$ uses that same jump with $\chi^{nn}_{mm} = \chi_{ee2}k^2$. Polynomial magnetic dispersion $\chi_{mm0} + \chi_{mm2}k_t^2$ is a centered second difference of the appended tangential H on the electric jump. Both polarizations carry it. A neighbor off the cut, or on the PEC wall, is refused when $\chi_{mm2} \neq 0$. A nonzero ξ_{mm2} appends one auxiliary Q per cut. The electric jump uses Q in place of that second difference, and $(1 - \xi_{mm2}\partial_{xx})Q$ equals the numerator on $H^- + H^+$. Both polarizations carry the auxiliary. The same neighbor is refused when $\xi_{mm2} \neq 0$. Every $\xi_{mm2} = 0$ leaves the polynomial stencil unchanged. The resistive conductance is a separate update. An embedded PMC is a mask on magnetic samples inside the grid. The outer wall is still `Boundary.PEC`, `Boundary.PERIODIC`, or `Boundary.PERIODIC_X`.

2.1 Staircase media

A sample is inside a region when its own coordinate is inside. Electric and magnetic samples are tested separately. There is no volume fraction and no conformal cut. A slab perpendicular to y occupies the half-open interval $y_0 \le y < y_1$. A disk of radius R centered at (x_c, y_c) occupies $(x - x_c)^2 + (y - y_c)^2 \le R^2$. Later regions overwrite earlier ones. The background is the law passed to `StaircaseIsotropic`, usually `UniformIsotropic` with vacuum values.

2.2 Bulk dispersion

One isotropic electric pole may fill the whole domain. The staircase does not yet vary the pole in space, and there is no magnetic dispersion.

The Debye permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \Delta\varepsilon}{1 + j\omega\tau}, \tag{3}$$

$$\tau\, \partial_t P + P = \varepsilon_0 \Delta\varepsilon\, E. \tag{4}$$

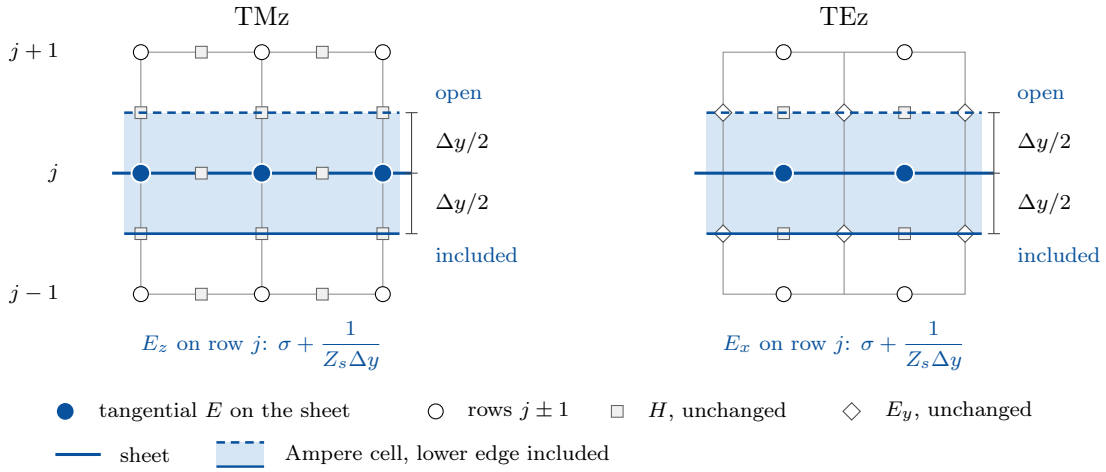

Figure 1: Where the resistive sheet enters the Yee cell. The blue line is the sheet, through tangential electric row j. The shaded band is the Ampere cell of those samples, $y_j - \Delta y/2 \le y < y_j + \Delta y/2$. It takes $\Delta y/2$ from each of the two primary cells that meet on the sheet. Every E_z on that row (TMz), and every E_x on that row (TEz), receives an added conductivity $1/(Z_s \Delta y)$. Rows $j-1$ and $j+1$ keep the conductivity of the volumetric law. Magnetic samples and the normal component E_y are unchanged. The upper edge is open, so a sheet that lands on it is added to the cell above. The lower edge belongs to the shaded cell.

The Lorentz permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 + j\gamma\omega}, \tag{5}$$

$$\partial_t^2 P + \gamma\, \partial_t P + \omega_0^2 P = \varepsilon_0 \omega_p^2 E. \tag{6}$$

These are bulk oscillators. They are not the surface oscillators in the GSTC section of the formulation note. Ampere's update subtracts the polarization current J_p with the impressed current,

$$\mathbf{E} \leftarrow C_a \mathbf{E} + C_b\big(\nabla \times \mathbf{H} - \mathbf{J}_p - \mathbf{J}\big). \tag{7}$$

C_a and C_b are the usual lossy coefficients built on ε_∞ and σ. The Lorentz auxiliary stores P at integer steps and advances it explicitly from E^n. The Debye auxiliary is centered, $E^{n+1/2} = (E^{n+1} + E^n)/2$, and the update is local to each sample. TEz keeps one auxiliary on E_x and one on E_y. A complex ε is rejected by the leapfrog. Passing ω into `sample_tm` or `sample_te` is how FDFD sees (3) or (5) directly.

3 Discrete method

3.1 Grid

The domain is $[0, n_x \Delta x] \times [0, n_y \Delta y]$. Array axis 0 is x and axis 1 is y, so a sample is indexed $[i, j]$. Locations are those of the Yee cell. For a PEC grid the boundary electric samples are stored. TMz places E_z on nodes, H_x on the y-edges, and H_y on the x-edges. TEz places H_z at cell centers, E_x on y-edges, and E_y on x-edges. E_x is zero on $y \in \{0, b\}$ and E_y is zero on $x \in \{0, a\}$ unless a Dirichlet trace replaces those zeros. Periodic identification drops the duplicate far side, and every stored sample is an unknown.

`YeeGrid2D.coordinates` returns the 1D sample coordinates of one component. Those coordinates are the positions the staircase tests.

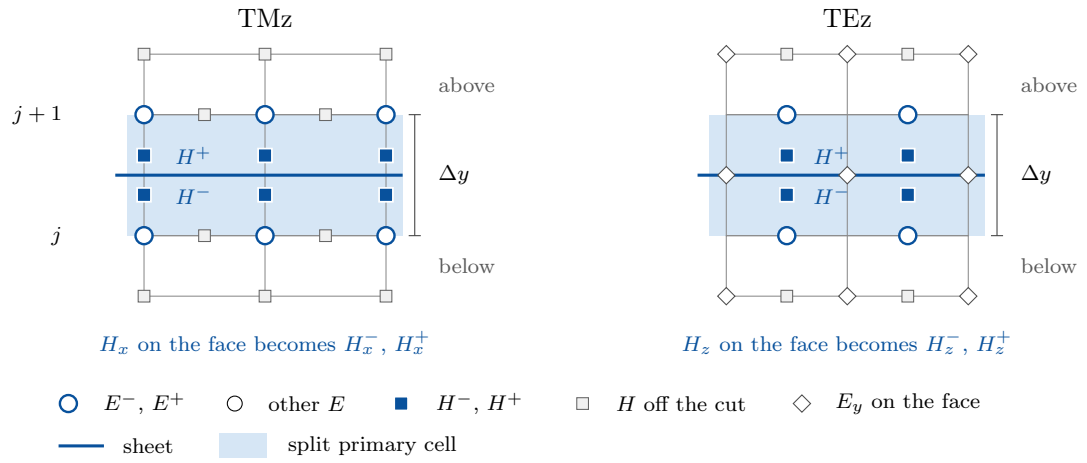

Figure 2: Where the symmetric sheet enters the Yee cell. The blue line is the sheet, on the magnetic face halfway between tangential electric rows j and $j+1$. The shaded band is that primary cell, of height Δy. The Yee sample that sat on the face is removed. TMz puts the pair H_x^-, H_x^+ in its place, and TEz puts H_z^-, H_z^+ there. The pair and the two electric rows close the jumps. Magnetic samples one cell off the cut, marked below and above, stay single Yee values. H_y stays on the electric rows. E_y stays on the face and receives the average of the pair. Rows j and $j+1$ keep the volumetric conductivity.

3.2 Curls

The discrete operators are centered differences. On the PEC grids they are transposes of each other once the boundary electric samples have been removed from the unknown vector. With phase one, the periodic curls are transposes as well. A Bloch phase makes `curl_h` the conjugate transpose of `curl_e`. An invalid degree-of-freedom id is -1 and must be masked before any component offset is added. TMz uses

$$(\nabla \times E)_x = \partial_y E_z, \qquad (\nabla \times E)_y = -\partial_x E_z, \tag{8}$$

and the matching Ampere pair. TEz uses

$$\nabla \times \mathbf{E} = \partial_x E_y - \partial_y E_x, \qquad E_x = \partial_y H_z, \qquad E_y = -\partial_x H_z. \tag{9}$$

The array curls and the sparse curls implement the same stencil. The array form keeps every sample, including the boundary values. The sparse pair used by FDFD drops the PEC rows.

The sign in (1) is not stored inside the factored FDFD matrix. Reconstructing a physical magnetic field from a Yee electric field requires

$$\mathbf{H} = -\frac{\nabla \times \mathbf{E}}{j\omega\mu}. \tag{10}$$

Omitting the minus negates H_x or H_z. The example plots use (10).

3.3 Leapfrog

One step advances $\mathbf{H}$ from time $n - \frac{1}{2}$ to $n + \frac{1}{2}$, then $\mathbf{E}$ from n to $n+1$. With real ε and σ,

$$C_a = \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)}, \qquad C_b = \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)}. \tag{11}$$

$J = 0$ and $J_p = 0$ is the milestone-1 update. The stable step on a uniform medium of speed c satisfies

$$\Delta t \le \frac{1}{c\sqrt{\Delta x^{-2} + \Delta y^{-2}}}. \tag{12}$$

When $\sigma = 0$ the leapfrog invariant, not the sum of the staggered sample energies, is the conserved quantity. The fully discrete frequency of a mode lies slightly above its semi-discrete frequency.

3.4 FDFD and the Dirichlet trace

Eliminating $\mathbf{H}$ produces the electric system

$$\nabla \times \left(\mu^{-1} \nabla \times \mathbf{E}\right) - \omega^2 \varepsilon \, \mathbf{E} = -j\omega \mathbf{J}, \tag{13}$$

stored as $A\mathrm{e} = -\mathrm{j}$. The real eigenproblem $K\mathrm{e} = \omega^2 M\mathrm{e}$ still requires real positive ε and μ and $\sigma = 0$. On a TEz PEC box the curl-curl operator has a gradient kernel, so a computed ω at roundoff is discarded before the physical spectrum is read.

Boundary electric samples stay out of the unknown vector. A trace e_D that equals the prescribed boundary values, and is zero on the free samples, moves to the right-hand side,

$$A_{ff}\, \mathrm{e}_f = -\mathrm{j} - (A e_D)_f. \tag{14}$$

The free-row action is the array form of the left-hand side of the FDFD equation. Calling `solve` with no trace, or with a zero trace, is the milestone-1 solve. The slab and the cylinder pass `dirichlet` set to the analytic boundary field and set $\mathrm{j} = 0$. This closes the box with a prescribed trace. The slab and the cylinder use that trace. The stretch that absorbs an outgoing wave is the next subsection.

3.5 Convolutional PML

An open side is a band of cells inside the PEC grid. The outer sample stays on the PEC wall, and depth ρ is measured from the inner face of the band toward that wall. The face itself, $\rho = 0$, is unstretched. On an axis w,

$$s_w = \kappa_w + \frac{\sigma_w}{\alpha_w + j\omega\varepsilon_0}, \tag{15}$$

and the centered difference that approximates ∂_w is multiplied by s_w^{-1} at the sample where that difference sits. Electric and magnetic samples lie half a cell apart, so the two curls carry different scales and the stretched pair is not a transpose. With every thickness zero the scale is absent, and the curls remain the real transpose of Sec. 3.2.

Inside a band of thickness d, with polynomial order m,

$$\sigma = \sigma_{\max} \left(\frac{\rho}{d}\right)^m, \qquad \sigma_{\max} = -\frac{(m+1)\ln R(0)}{2\eta_0 d}, \tag{16}$$

together with $\kappa = 1 + (\kappa_{\max} - 1)(\rho/d)^m$ and $\alpha = \alpha_{\max}(1 - \rho/d)^m$. The defaults are $m = 3$, $R(0) = 10^{-6}$, $\kappa_{\max} = 1$, and $\alpha_{\max} = 0$. Each side uses its own thickness, the cell count times the spacing on that axis. A node on the PEC wall has $\rho = d$, so $\sigma = \sigma_{\max}$, $\kappa = \kappa_{\max}$, and $\alpha = 0$. `PMLSpec.box` puts the same cell count on all four sides. A periodic grid, and the periodic-x wall, are rejected: the band needs the PEC backing.

FDFD rebuilds the sparse curls at the solve frequency, since s_w depends on ω. The eigenproblem keeps the unstretched real curl, and `spatial_operator` refuses an active PML. It also refuses a nonzero Bloch wave number, because that phase makes the curls complex.

The leapfrog stores one auxiliary per stretched derivative. It advances that auxiliary with the exponential integrator of $1/s_w$, on the same Δt as the field step. C_a and C_b stay on the material samples. A Debye or Lorentz callback still receives $\nabla \times \mathbf{H}$ after the magnetic update, and that curl is the stretched one. With no PML object, both drivers stay on the milestone-1 update.

3.6 Embedded conductors

PEC and PMC are sample masks. Setting ε to infinity, or μ to zero, is a different model. PEC removes every electric sample whose own coordinate lies in the object. PMC removes every magnetic sample in its region. The leapfrog writes those samples back to zero: PEC electric samples together with the outer wall, and PMC magnetic samples after the magnetic update, before Ampere's law reads them. The eigenproblem uses that reduced real curl. It still refuses an active PML.

The regions use the staircase tests of Sec. 2.1. A band is the half-open interval $y_0 \le y < y_1$. A circle is closed, so a sample on the circle is inside. The outer wall stays `Boundary.PEC`, `Boundary.PERIODIC`, or `Boundary.PERIODIC_X`.

The open PEC cylinder solves for the scattered field on the PML of Sec. 3.5. The trace in (14) equals $-E_{\text{inc}}$ on the object and zero on the outer wall. A free sample of the total field is $E_s + E_{\text{inc}}$. Inside the object the total electric field is zero. The TEz incident wave is $H_z = e^{-jk(x-x_c)}$, with $E_x = 0$ and $E_y = (k/\omega\varepsilon_0)\, H_z$.

3.7 Analytic checks

Both scattering checks use the trace (14) on every PEC electric sample. The interior material is the staircase of Sec. 2.1. The comparison is the gap left by numerical dispersion and, for the cylinder, by the staircase.

Fresnel slab. A non-magnetic slab $y_1 \le y < y_2$ of index $n = \sqrt{\varepsilon_r}$ sits in vacuum. The incident wave travels toward $+y$ and has phase zero at the front face. TMz reflects with $r = (1-n)/(1+n)$. TEz reflects H_z with $r_H = (n-1)/(n+1) = -r$, so $R_H = -R$ while the transmission amplitude, including the transit phase e^{-jkd}, is the same. The interior amplitudes are not the same: TMz splits $1 \pm R$ with $1/n$, and TEz splits $1 \pm R_H$ with n. When $\varepsilon_r = 1$, both reflections vanish and the field is the incident wave, not a transmission of 1. The closed forms are `slab_coefficients` and `slab_fields`. A forced branch (`below`, `slab`, `above`) evaluates one piecewise piece at a face, where the two neighboring expressions must agree.

The comparison wavelength is kept off the TEz eigenfrequencies of the PEC box. On the 8×80 domain with $\Delta = 1$, $\lambda = 40$ is one of those frequencies and the Dirichlet matrix is singular. The shipped example uses $\lambda = 46$, $\Delta = 0.5$, and faces $y = 30.375$ and $y = 50.375$, which miss the electric nodes of both that grid and the once-refined grid.

Dielectric cylinder. The outgoing wave is $H_n^{(2)}$. The incident field e^{-jkx} has the Jacobi–Anger coefficients $(-j)^n$. TMz is the E_z series. TEz is the H_z series, with the contrast on the undifferentiated interior Bessel function, because E_φ brings $1/\varepsilon$. The PEC coefficient $a_n = -(-j)^n J_n(ka)/H_n^{(2)}(ka)$ is a separate regression. It is not the limit $\varepsilon_r \to \infty$ of the dielectric formula. The TEz form, from $E_\varphi = 0$, is $a_n = -(-j)^n J_n'(ka)/H_n^{(2)\prime}(ka)$. The truncation follows the MaxwellMOM rule $\lceil |ka| + 4|ka|^{1/3} + 8 \rceil$, and the derivative is the recurrence $\frac{1}{2}(f_{n-1} - f_{n+1})$.

The open dielectric cylinder compares this series on the PML of Sec. 3.5. The unknown is the scattered field and the outer wall is homogeneous. The incident wave enters as the impressed current $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$, sampled with the staircase permittivity that stays in the operator. A free sample of the total field is $E_s + E_{\text{inc}}$. The closed-box cylinder still prescribes the Mie trace.

Line current. A z-directed current whose integral is I satisfies $(\nabla^2 + k^2)E_z = j\omega\mu J_z$ in a uniform medium. For $e^{+j\omega t}$ the outgoing field is $E_z = -(\omega\mu I/4)\, H_0^{(2)}(k\rho)$, with $k = \omega\sqrt{\mu\varepsilon}$ taken from the same μ and ε the solver uses. The azimuthal field is $H_\varphi = -j(kI/4)\, H_1^{(2)}(k\rho)$.

The discrete sample is $J_z = I/(\Delta x \Delta y)$. The comparison keeps the unstretched samples at least $\lambda/2$ from the source. The continuous field diverges at $\rho = 0$, and that sample is left undefined.

Hertzian dipole. A z-directed element of moment $I\ell$ radiates the outgoing spherical wave for $e^{+j\omega t}$. Off the element, $H_\varphi = (jkI\ell/4\pi)\,(1 + 1/(jkr))\,e^{-jkr}\sin\theta/r$, $E_r = \eta I\ell\cos\theta\,(1 + 1/(jkr))\,e^{-jkr}/(2\pi r^2)$, and $E_\theta = j\eta kI\ell\sin\theta\,(1 + 1/(jkr) - 1/(kr)^2)\,e^{-jkr}/(4\pi r)$. The discrete sample is $J_z = I\ell/(\Delta x \Delta y \Delta z)$ on one E_z edge. The comparison keeps the unstretched electric samples at least $\lambda/4$ from the element. The continuous field diverges there.

Gaussian beam. A 2D paraxial beam of waist radius w_0 and amplitude A propagates in $+\hat{\mathbf{y}}$ for $e^{+j\omega t}$. With $z_R = kw_0^2/2$ and $q = z + jz_R$, $E_z = A\sqrt{jz_R/q}\,e^{-jkz - jkx^2/(2q)}$, and the local plane-wave magnetic field is $H_x = E_z/\eta$. The discrete source is the impressed current on the two E_z rows that touch a horizontal split. Rows below the split hold the scattered field. The comparison keeps unstretched total-field samples at least $\lambda/2$ from the split.

Near-to-far field. A closed contour in the unstretched region carries E_z and the tangential magnetic field. With $\mathbf{n}$ the outward unit normal and $K_z = (\mathbf{n} \times \mathbf{H})_z$, the far-zone field at distance ρ and angle φ from $+\hat{\mathbf{x}}$ is the leading Hankel factor $(j/4)\sqrt{2/(\pi k\rho)}\,e^{-j(k\rho - \pi/4)}$ times the contour integral of $e^{jk\hat{\boldsymbol{\rho}}\cdot\mathbf{r}'}\big[j\omega\mu K_z - jk(\mathbf{n}\cdot\hat{\boldsymbol{\rho}})E_z\big]$. On a Yee rectangle the electric field is the node value and the tangential H is the average of the two faces that touch the node line. The magnetic field on that contour comes from the unstretched curl, $H = -(\nabla \times E)/(j\omega\mu)$, so the rectangle stays off the PML. The same Hankel factor, with amplitude $-(\omega\mu I/4)$, is the far field of a line current at the origin. The comparison uses that wave.

3.8 3D cell

The 3D grid stores all six components. E_x sits at $((i + \frac{1}{2})\Delta x,\, j\Delta y,\, k\Delta z)$, and the other electric components are staggered on their own axes. PEC drops the tangential electric samples. Magnetic samples that do not touch a free electric sample are omitted, so the unstretched curls remain transposes. The FDFD matrix is the elimination in (13), with the diagonal $\varepsilon + \sigma/(j\omega)$ when ε is real. The rectangular cavity frequency is the two-dimensional semi-discrete formula with the added term $(2\sin(k_z\Delta z/2)/\Delta z)^2$ and $k_z = p\pi/c$. A mode with exactly one index equal to zero is one Cartesian component: $(m, n, 0)$ is E_z, $(m, 0, p)$ is E_y, and $(0, n, p)$ is E_x.

A staircase ball is closed. A sample is inside when $(x - x_c)^2 + (y - y_c)^2 + (z - z_c)^2 \le a^2$, and electric and magnetic samples are tested separately. The 3D FDFD operator can rebuild every partial derivative with the stretch $1/s_w(\omega)$ on a band inside the PEC wall. The real eigenproblem refuses that stretch. A 3D leapfrog is not in this slice. The open dielectric sphere uses the contrast current of the open cylinder: the ball stays in the diagonal, the unknown is the scattered field, and a free sample of the total field is $E_s + E_{\text{inc}}$. The incident wave travels toward $+z$ and equals $\hat{\mathbf{x}}$ at the sphere center.

4 Code

The installable package is `maxwell_fd`, version 0.1.0. Its source is `src/maxwell_fd`. NumPy and SciPy are required. Matplotlib is the `viz` extra and is also a dependency of the conda environment `maxwell-fd`.

Module	Role
`grid.yee2d`	`YeeGrid2D`, `Polarization`, `Boundary`, coordinates
`materials.volume`	Uniform, staircase, mapped $\varepsilon(x, y)$, component arrays
`materials.dispersion`	`DebyeDielectric`, `LorentzDielectric`, and their auxiliaries
`materials.conductors`	Staircase PEC and PMC masks, `YBand` and `Circle`
`operators.curl2d`	Sparse `curl_e`, `curl_h`, array curls, DOF layout, optional stretch, Bloch phase on the x wrap
`operators.gstc`	Symmetric cell: remove the cut sample and append H^-, H^+
`operators.pml`	CFS profile, $1/s_w(\omega)$, the 2D leapfrog auxiliaries, and the 3D FDFD stretch
`drivers.fdtd2d`	`step_tmz`, `step_tez`, CFL step, leapfrog invariant
`drivers.fdfd2d`	`FDFDOperator`: system, eigenproblem, Dirichlet solve, optional sheet
`analytics.cavity2d`	Semi-discrete cavity frequencies and mode shapes
`analytics.dispersion`	Fully discrete frequency and the three-sample fit
`analytics.fresnel`	Slab coefficients and fields
`analytics.mie2d`	Cylinder coefficients and E_z, H_z, E_x, E_y
`analytics.green2d`	Outgoing line-current E_z and H_φ
`analytics.farfield2d`	2D TMz near-to-far integral
`grid.yee3d`	`YeeGrid3D`: six components, PEC and periodic shapes
`materials.volume3d`	Uniform isotropic law and a staircase ball
`operators.curl3d`	Sparse and array curls on the 3D cell, optional $1/s_w$
`drivers.fdfd3d`	`FDFDOperator3D`: system, real eigenproblem, optional PML
`analytics.cavity3d`	Semi-discrete frequency with the z term, and the pure modes
`analytics.mie3d`	Sphere fields with the outgoing wave $h_n^{(2)}$
`materials.sheets`	`ResistiveSheet` on one Ampere cell, and `SymmetricSheet` with piecewise $\chi(x)$
`analytics.sheets`	Sheet R, T, and the uniform and oblique fields

A frequency-domain TMz solve with a staircase disk has this shape. The trace argument is omitted when the wall is a homogeneous PEC.

```
grid = YeeGrid2D(nx, ny, dx, dy, Polarization.TMZ, Boundary.PEC)
law = StaircaseIsotropic(
    UniformIsotropic(),
    (Disk(cx, cy, radius, eps=eps_r * EPS0),),
)
operator = FDFDOperator(grid, law.sample_tm(grid))
ez = operator.solve(omega, np.zeros(operator.layout.n_e),
                    dirichlet={"ez": ez_trace})
```

`sample_tm(grid)` and `sample_te(grid)` without ω return the real instantaneous coefficients used by the leapfrog (ε_∞ when the law is dispersive). Passing ω allows a complex permittivity. `step_tmz` and `step_tez` take the impressed current and the polarization current as optional arguments. Their defaults reproduce the milestone-1 update. The same two steps take an optional `ConvolutionalPML` and an optional `Conductors` mask. `FDFDOperator` takes an optional `PMLSpec` and the same mask. Passing neither a PML nor a conductor leaves both drivers on the full unstretched curls. A resistive sheet is applied to the sampled component arrays before either driver is built. It adds $1/(Z_s \Delta y)$ to σ and does not change the solve or the step. A symmetric sheet is a keyword of `FDFDOperator`. It appends H^- and H^+ and returns the electric field. It does not paint σ. χ_{mm}^{nn} adds $\partial_x H_y$ on the TMz jump. The TEz rows stay the tangential pair. A finite run passes the ends of the half-open interval. An incident wave passed to `solve` makes the unknown the scattered field. The sheet may share a PML box when every cut sample is unstretched. A piecewise sheet passes one interval on each piece, and each cut sample uses the

susceptibilities of the piece that contains its x. Polynomial electric dispersion passes χ_{ee0} and $\chi_{mm}^{nn} = \chi_{ee2}k^2$ to that same sheet. Polynomial magnetic dispersion stores χ_{mm2} and differences the appended tangential H on the electric jump, in both polarizations. A nonzero ξ_{mm2} appends one auxiliary per cut and replaces that difference.

The real eigenproblem is `spatial_operator`. It returns the pair (K, M) of the unstretched curl and refuses an active PML. TEz results should drop nonpositive eigenvalues before they are compared with the semi-discrete formula.

5 Examples

Scripts under `examples/fdfd/2d/`, `examples/fdfd/3d/`, and `examples/fdtd/` are the runnable demos. The time-domain tree does not yet solve a scattering problem. From the project directory, with the conda environment `maxwell-fd` active,

```
python examples/fdfd/2d/fresnel_slab.py
python examples/fdfd/2d/dielectric_cylinder.py
python examples/fdfd/2d/pml_line.py
python examples/fdfd/2d/pec_cylinder.py
python examples/fdfd/2d/dielectric_cylinder_open.py
python examples/fdfd/2d/resistive_sheet.py
python examples/fdfd/2d/symmetric_sheet.py
python examples/fdfd/2d/finite_sheet.py
python examples/fdfd/2d/piecewise_sheet.py
python examples/fdfd/2d/dispersive_sheet.py
python examples/fdfd/2d/magnetic_dispersive_sheet.py
python examples/fdfd/2d/magnetic_rational_sheet.py
python examples/fdfd/2d/periodic_sheet.py
python examples/fdfd/3d/cavity.py
python examples/fdfd/3d/dielectric_sphere.py
```

Each script writes a sibling directory whose name is the script stem plus the parameters of that run, in the order they are passed to `examples/run_dir.output_dir`. Floats are formatted with twelve significant digits and trailing zeros removed, so `2.0` becomes `2` and `0.5` stays `0.5`. Keys do not contain underscores.

Field maps use `pcolormesh` with Gouraud shading. Each polarization is a row of three columns: the Yee field, the analytic field, and their difference. The cylinder columns are FD, Mie, and FD−Mie. The slab columns are FD, Fresnel, and FD−Fresnel. The line-current columns are FDFD, Hankel, and FDFD−Hankel. Color limits of the first two columns match. The difference has its own scale. The cylinder outline, the slab faces, or the inner edge of the PML, is drawn on every panel. The open PEC cylinder and the open dielectric cylinder draw the outline and that inner edge together. The 3D cavity figure is one panel per pure mode. The open sphere is one row of $\mathrm{Re}\, E_x$ on the xz plane through the center: FD, Mie, and FD−Mie. The resistive sheet and the symmetric sheet use the columns FD, Sheet, and FD−Sheet, with $\mathrm{Re}\, E_z$ and $\mathrm{Re}\, E_x$. The periodic oblique sheet, the dispersive sheet, the magnetic dispersive sheet, and the rational magnetic sheet use FD, Analytic, and FD−Analytic. The finite sheet and the piecewise sheet use Incident, Scattered, and Total.

5.1 Fresnel slab

```
python examples/fdfd/2d/fresnel_slab.py \
    --index 2 --thickness 20 --y1 30.375 --wavelength 46 \
    --length 8 --height 80 --dx 0.5 --pml 1
```

`-index` is the refractive index n, and the permittivity used in the solve is $\varepsilon_r = n^2$. The default $n = 2$ is $\varepsilon_r = 4$. `-length` and `-height` are the region sides along x and y. Both must be a positive integer number of cells of size `-dx`, and each direction needs at least two cells. `-pml` is a thickness at each end in y only. It must be a positive multiple of Δ and must leave the slab in the open interval (`pml`, `height` $-$ `pml`).

The default run writes

```
fresnel_slab_index2_d20_lam46_y130.375_length8_height80_dx0.5_pml1/
```

- `domain.png`, the mesh, the slab, the y-end bands, and the x walls;
- `fresnel_slab.png`, TMz E_z and TEz H_z in three columns;
- `cuts.png`, the center-column cuts of E_z, H_x, E_x, and H_z against the analytic curves.

On this grid the free-sample electric L^2 errors annotated on the figures are 0.23% (TMz) and 1.30% (TEz). Those numbers are the result of this particular grid. The acceptance bars in the formulation note are wider and were not changed to match them.

Propagation is $+y$, drawn horizontal so the figure reads with the line cuts. The incident field does not depend on x, and neither does the slab. The orange bands are therefore only the two ends $y = 0$ and $y = b$. The walls $x = 0$ and $x = a$ are drawn in green and labeled as the PEC trace: every boundary electric sample on those walls is set to the analytic Fresnel value, together with the samples on $y = 0$ and $y = b$. The slab rectangle spans the full width in x. Figure 3 is that layout. The orange bands mark a drawn thickness at $y = 0$ and $y = b$. `FDFDOperator` is constructed here without a `PMLSpec`, so every derivative stays the centered difference of Sec. 3.2 and the trace is the analytic field. The stretch itself is the line-current example of Sec. 5.3.

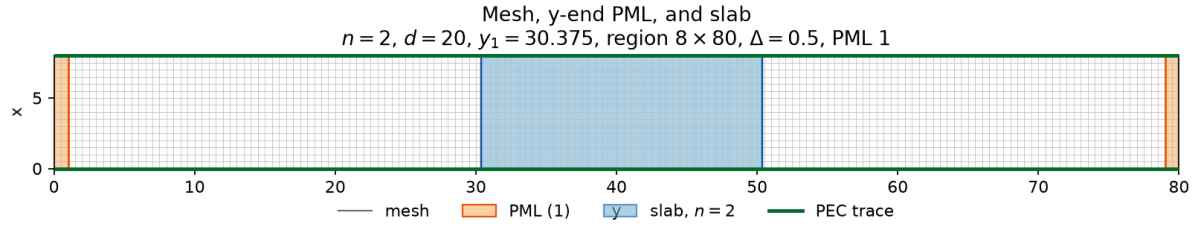

Figure 3: Default Fresnel domain. Orange bands are the y-end PML thickness drawn on the figure. Green edges are the PEC trace at $x = 0$ and $x = 8$.

5.2 Dielectric cylinder

```
python examples/fdfd/2d/dielectric_cylinder.py \
    --radius 10 --index 1.5 --size 50 --dx 1 --pml 8
```

The region is a square of side `-size`. `-index` is again n, with $\varepsilon_r = n^2$. The default $n = 1.5$ is $\varepsilon_r = 2.25$. The size parameter ka is held at 1, so $k = 1/a$ and the wavelength scales with the radius. The radius must be positive and smaller than half the side length minus the PML thickness. The default run writes

```
dielectric_cylinder_index1.5_ka1_a10_size50_dx1_pml8/
```

with `domain.png` and `dielectric_cylinder.png`.

The cylinder scatters in the plane of the grid, so the domain figure draws the PML thickness on all four sides. As in the slab example, that frame is a layout. The solve is still (14) on the PEC wall, with the Mie field as the trace and with the staircase disk as the only material contrast. Every derivative in this solve is the centered difference of Sec. 3.2. TMz shows $\mathrm{Re}\, E_z$. TEz shows $\mathrm{Re}\, H_z$ from (10), and the annotated L^2 error is the free-sample error in E_x and E_y. On the default grid those errors are 0.44% and 2.07%.

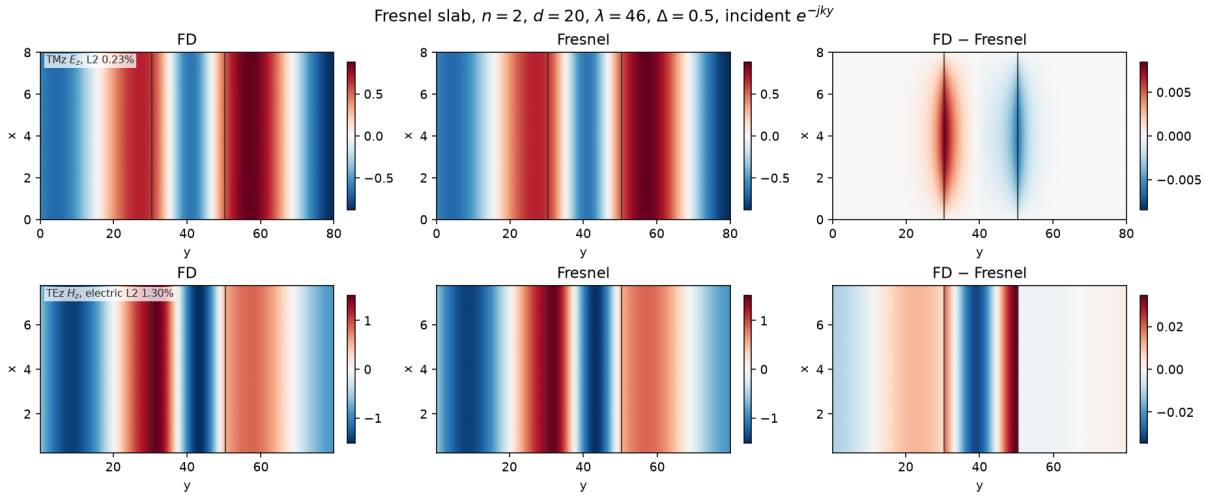

Figure 4: Default Fresnel maps. Vertical lines mark the slab faces. H_z is rebuilt with (10). The annotated L^2 errors are the free electric samples.

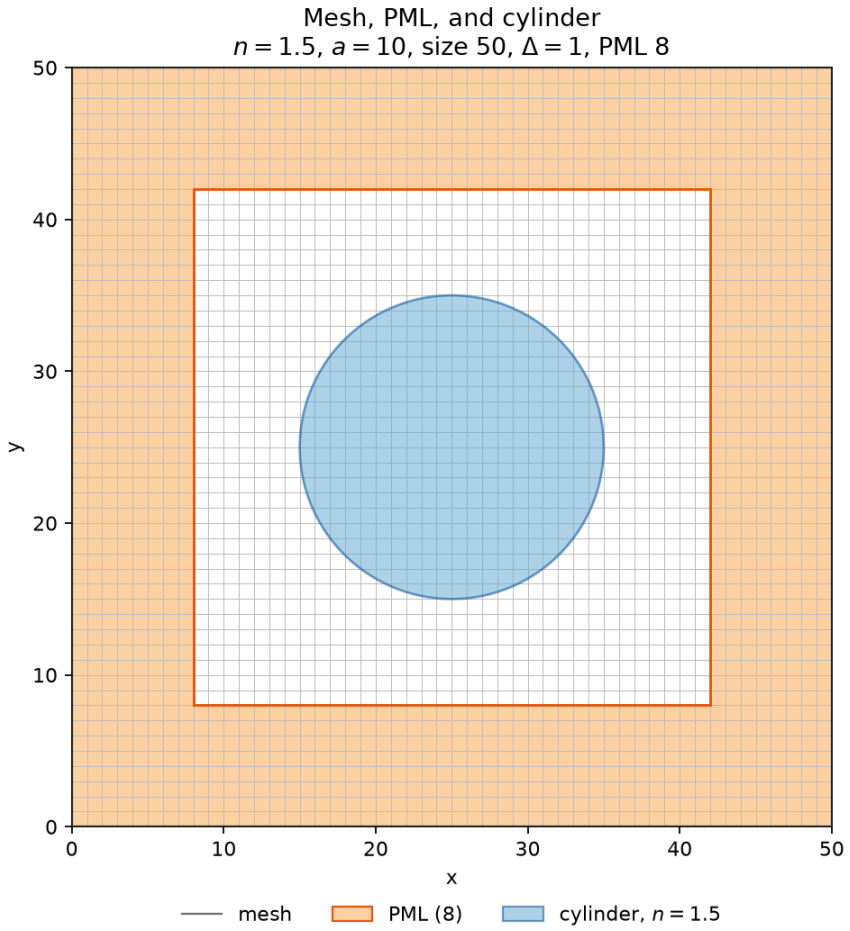

Figure 5: Default cylinder domain. The orange frame is the drawn PML thickness. The disk is the staircase geometry. The outer boundary of the solve remains the PEC trace.

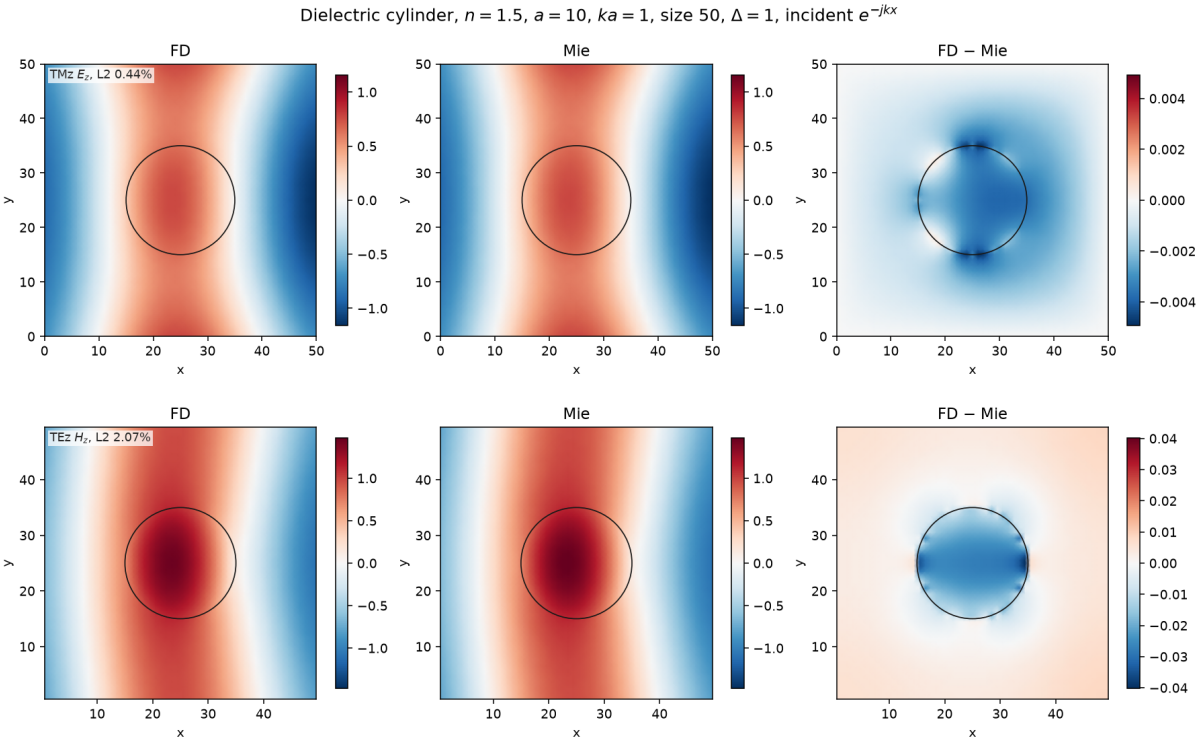

Figure 6: Default cylinder maps against the Mie series at $ka = 1$, $n = 1.5$.

5.3 Line current in a PML

```
python examples/fdfd/2d/pml_line.py \
    --wavelength 1 --side 3 --dx 0.05 --pml 8
```

The domain is a square PEC box of side `-side`. `-dx` is the cell size, and `-side` must be an integer number of those cells, at least two. `-pml` is the cell count on every side. It is at least one, and twice that count stays below the number of cells, so an interior remains. `-wavelength` is the free-space wavelength. The source is the center E_z sample, $J_z = 1/(\Delta x\, \Delta y)$, and the reference is the line current of Sec. 3.7 with $I = 1$. The default run writes

`pml_line_lam1_side3_dx0.05_pml8/`

and `pml_line.png`. The orange square is the inner edge of the band. On this grid the interior L^2 error, outside half a wavelength of the source, is 2.10%. That number belongs to this grid. The Hankel panel is the unbounded cylindrical wave, including through the band, so the difference outside the square is the absorption.

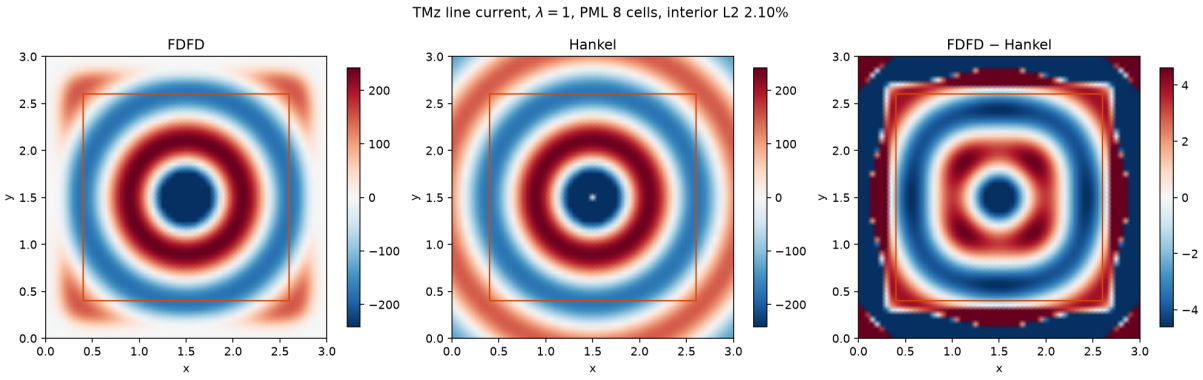

Figure 7: Line current at $\lambda = 1$ on a 3 m box, $\Delta = 0.05$ m, with eight PML cells on each side. The orange square is the inner edge of the band. The annotated error is the interior window described above.

5.4 Open PEC cylinder

```
python examples/fdfd/2d/pec_cylinder.py \
    --radius 8 --size 64 --dx 1 --pml 8
```

The region is a square of side `-size`. `-pml` is the cell count of the convolutional PML on each side, the stretch of Sec. 3.5. ka stays at 1. The solve frequency is $\omega = c_0/a$, and the Mie series uses $k = \omega\sqrt{\mu_0\varepsilon_0}$. The radius must lie inside the unstretched square. The default run writes

```
pec_cylinder_ka1_a8_size64_dx1_pml8/
```

with `domain.png` and `pec_cylinder.png`.

The unknown is the scattered field. The cylinder trace is $-E_{\text{inc}}$ and the outer wall is zero. A free sample of the total field is $E_s + E_{\text{inc}}$. TMz shows $\operatorname{Re} E_z$. TEz shows $\operatorname{Re} H_z$ from (10). The annotated L^2 is the electric-sample error on the unstretched samples at least 2Δ outside the circle. On this grid those errors are 4.20% (TMz) and 5.58% (TEz). The orange square is the inner edge of the PML. Outside that square the difference is the absorption together with the scattered field's wall condition.

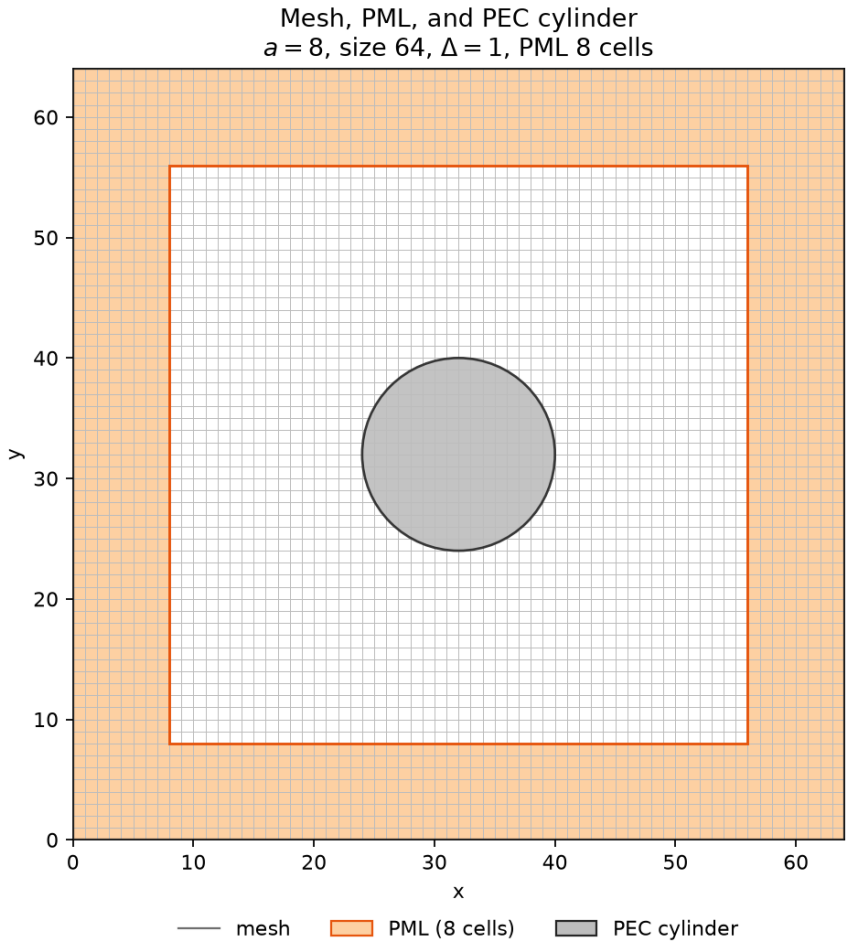

Figure 8: Default open-cylinder domain. The orange frame is the PML band used by the solve. The disk is the staircase PEC.

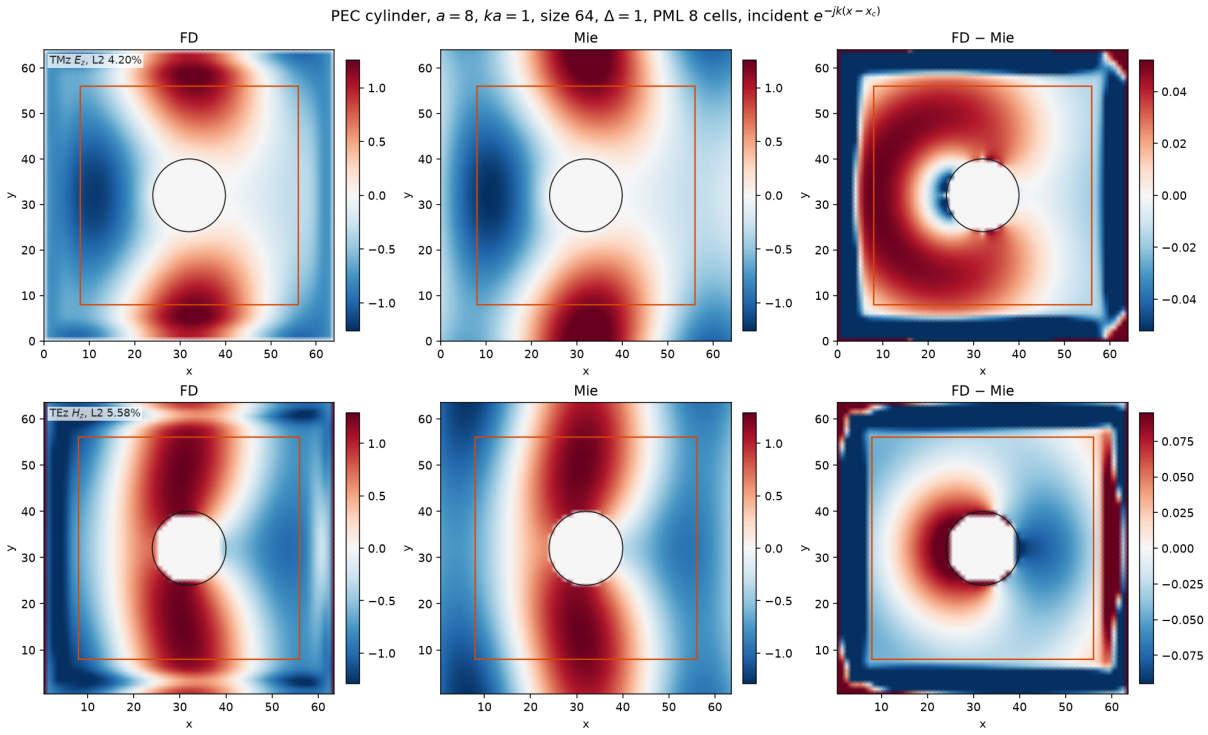

Figure 9: Open PEC cylinder at $ka = 1$, $a = 8$, $\Delta = 1$, with eight PML cells on each side. The annotated L^2 errors are the electric samples in the unstretched window, at least 2Δ outside the circle.

5.5 Open dielectric cylinder

```
python examples/fdfd/2d/dielectric_cylinder_open.py \
    --radius 8 --index 1.5 --size 64 --dx 1 --pml 8
```

The region is a square of side `-size`. `-index` is n, with $\varepsilon_r = n^2$. The default $n = 1.5$ is $\varepsilon_r = 2.25$. `-pml` is the cell count of the convolutional PML on each side, the stretch of Sec. 3.5. ka stays at 1. The solve frequency is $\omega = (ka/a)\, c_0$, and the Mie series uses $k = \omega\sqrt{\mu_0 \varepsilon_0}$. The radius must lie inside the unstretched square. The default run writes

```
dielectric_cylinder_open_index1.5_ka1_a8_size64_dx1_pml8/
```

with `domain.png` and `dielectric_cylinder_open.png`.

The unknown is the scattered field. The operator holds the staircase disk. The impressed current is $J = j\omega(\varepsilon - \varepsilon_0)E_{\text{inc}}$ on the electric samples, and it is zero in the vacuum. The outer wall is a homogeneous PEC. A free sample of the total field is $E_s + E_{\text{inc}}$. TMz shows $\operatorname{Re} E_z$. TEz shows $\operatorname{Re} H_z$ from (10). The annotated L^2 is the electric-sample error on the unstretched samples at least 2Δ off the circle, inside and outside. On this grid those errors are 0.45% (TMz) and 0.71% (TEz). The orange square is the inner edge of the PML. The closed-box cylinder of Sec. 5.2 still prescribes the Mie trace and does not stretch the derivatives.

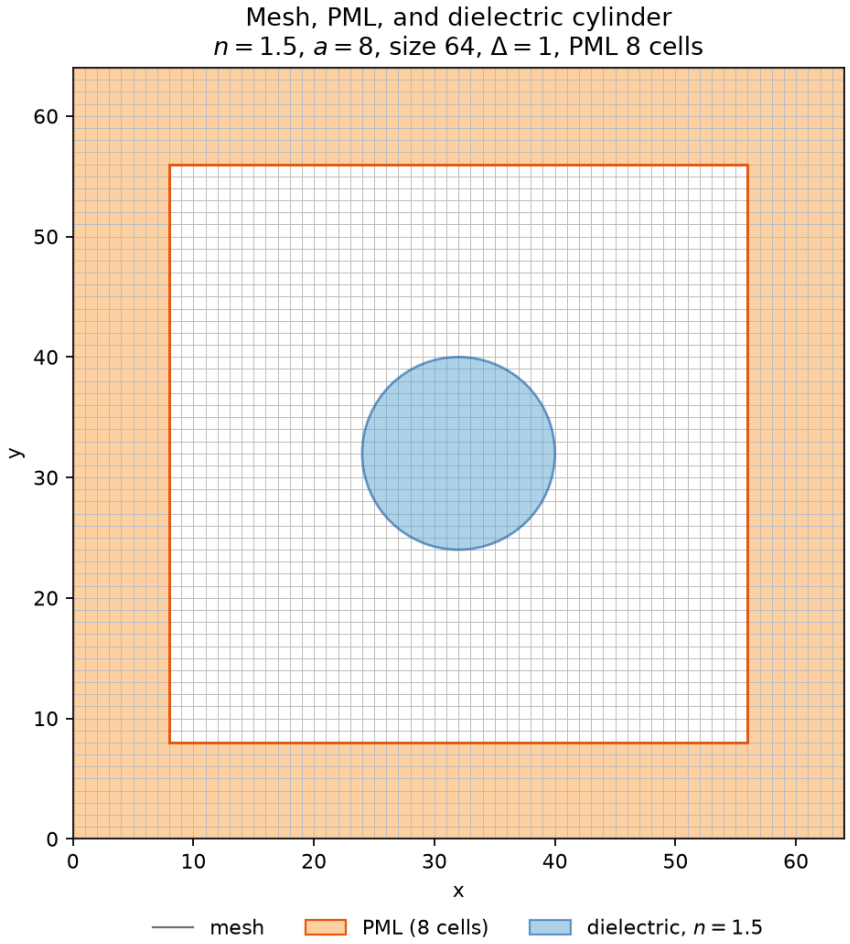

Figure 10: Default open dielectric domain. The orange frame is the PML band used by the solve. The disk is the staircase dielectric, $n = 1.5$.

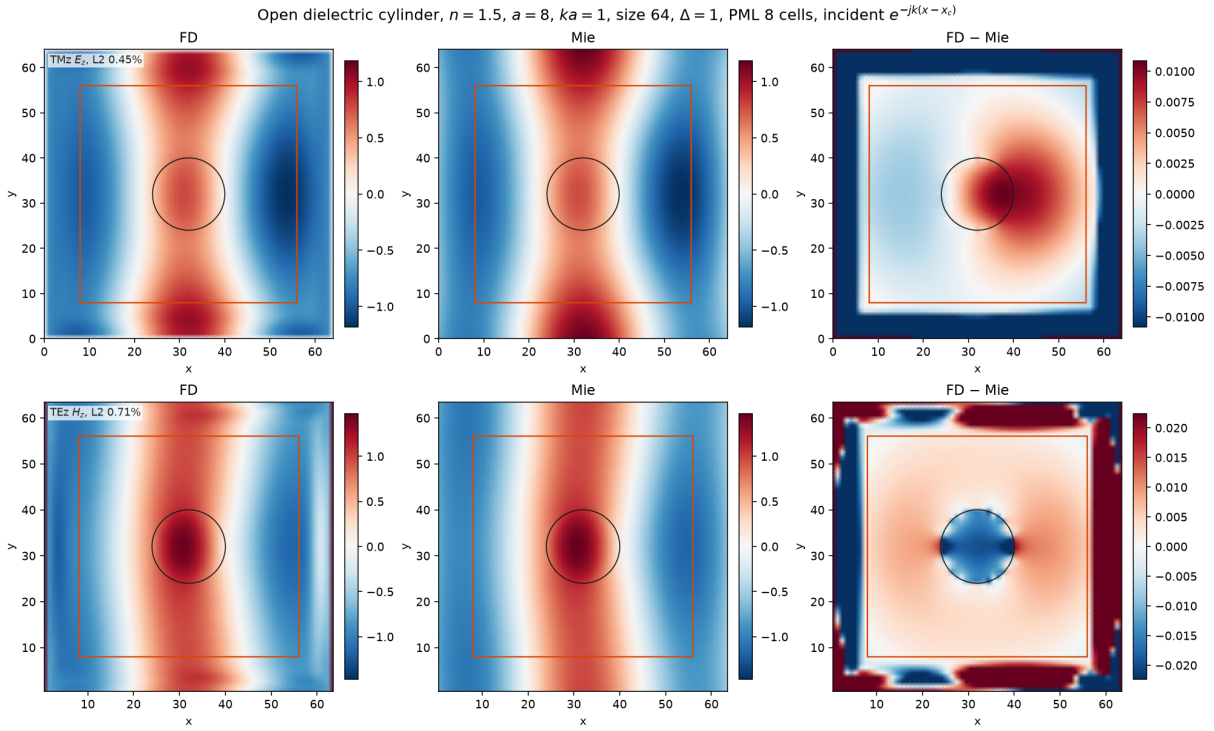

Figure 11: Open dielectric cylinder at $n = 1.5$, $ka = 1$, $a = 8$, $\Delta = 1$, with eight PML cells on each side. The annotated L^2 errors are the electric samples in the unstretched window, at least 2Δ off the circle.

5.6 3D PEC cavity

```
python examples/fdfd/3d/cavity.py --nx 16 --ny 12 --nz 10 --dx 1
```

The box is a PEC cavity with one spacing on every axis. The default run writes

```
cavity_nx16_ny12_nz10_dx1/
```

with `cavity.png`. The three panels are the pure modes $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$. Each panel shows the one nonzero electric component, on the plane in which that mode varies. The script prints the relative Rayleigh residual of each mode on that grid. The regression in the test suite uses the coarser $8 \times 6 \times 5$ box and the bar 10^{-10}.

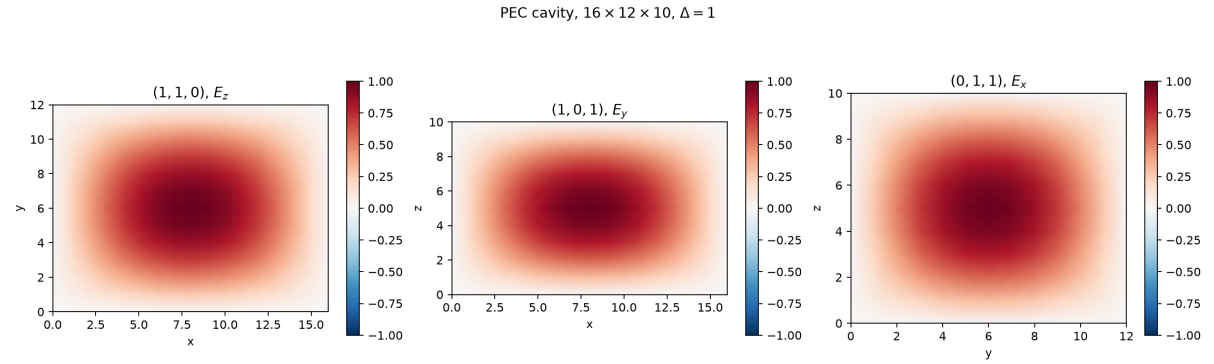

Figure 12: Pure modes of a $16 \times 12 \times 10$ PEC cavity at $\Delta = 1$. $(1, 1, 0)$ is $E_z(x, y)$, $(1, 0, 1)$ is $E_y(x, z)$, and $(0, 1, 1)$ is $E_x(y, z)$.

5.7 Open dielectric sphere

```
python examples/fdfd/3d/dielectric_sphere.py \
    --radius 4 --index 1.5 --size 18 --dx 1 --pml 3
```

The default run writes

```
dielectric_sphere_index1.5_ka1_a4_size18_dx1_pml3/
```

with `domain.png` and `dielectric_sphere.png`. The cube is 18 cells on a side at $\Delta = 1$, the radius is 4, the index is 1.5, and $ka = 1$. Three PML cells back every face. The unknown is the scattered field. The staircase ball stays in the operator, and the incident wave $\hat{\mathbf{x}}\, e^{-jk(z-z_c)}$ enters as the contrast current. The figure shows $\mathrm{Re}\, E_x$ on the xz plane through the center. The annotated L^2 is the electric-sample error on all three components, outside the PML and at least 2Δ off the surface. On this grid that error is 0.98%. The regression uses the same grid and the bar 1.8×10^{-2}.

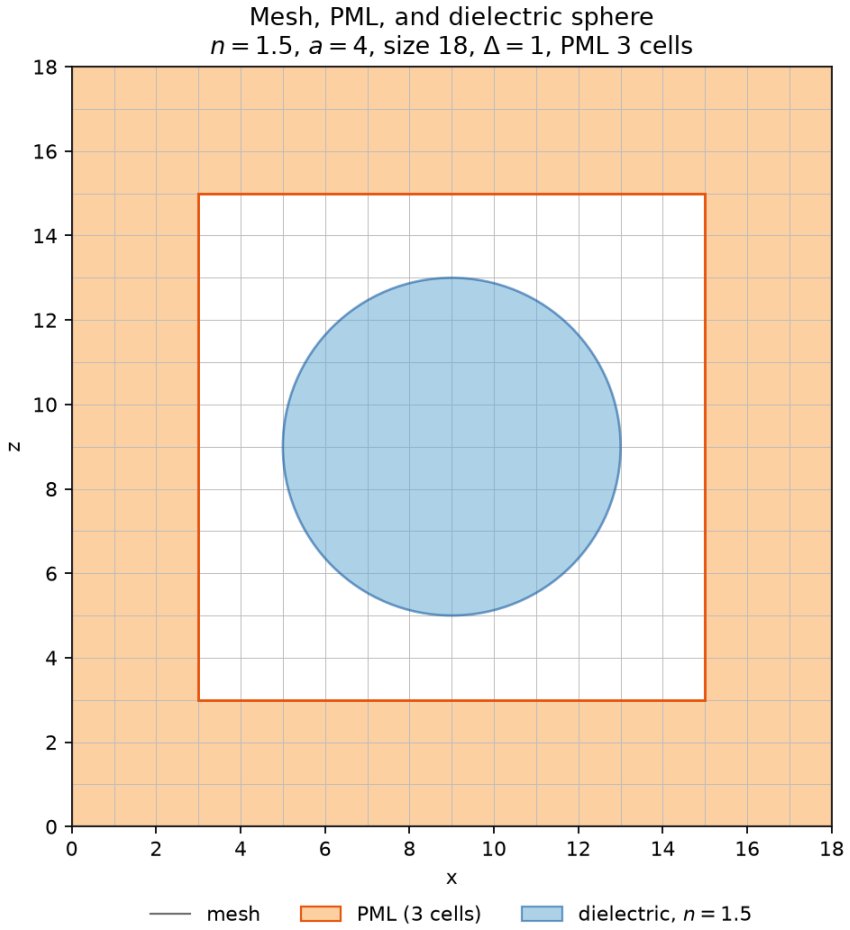

Figure 13: Default open-sphere domain. The orange frame is the PML band used by the solve. The disk is the cross-section of the staircase ball, $n = 1.5$.

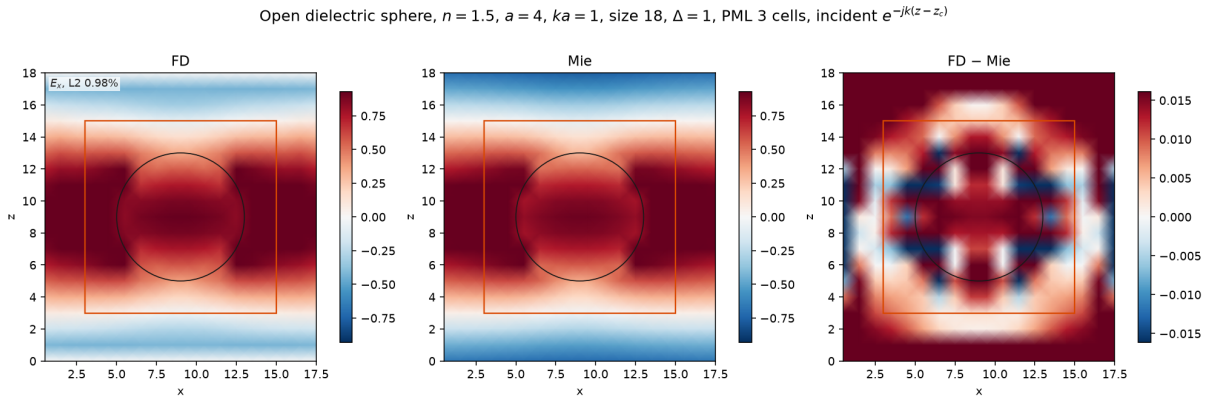

Figure 14: Open dielectric sphere at $n = 1.5$, $ka = 1$, $a = 4$, on an 18^3 box at $\Delta = 1$, with three PML cells on each face. The panels are $\mathrm{Re}\, E_x$ on the xz plane through the center. The annotated L^2 is the electric-sample error outside the PML and at least 2Δ off the surface.

5.8 Hertzian dipole

```
python examples/fdfd/3d/hertzian_dipole.py
```

The element is z-directed with moment $I\ell = 1$. It sits on the E_z sample nearest the center of a PEC cube, and that sample carries $J_z = 1/(\Delta x \Delta y \Delta z)$. `-wavelength`, `-size`, `-dx`, and `-pml` are the free-space wavelength, the cube side, the cell size, and the PML depth on each face. The defaults are 1, 1.6, 0.1, and 3. The comparison is every free electric sample outside the PML and at least 0.25λ from the element.

The default run writes

```
hertzian_dipole_lam1_size1.6_dx0.1_pml3/
```

with `domain.png` and `hertzian_dipole.png`. The cube is 16 cells on a side. Three PML cells back every face. The figure shows $\mathrm{Re}\, E_z$ on the xz plane through the element. On this grid the annotated L^2 is 9.65%. The regression uses the same grid and the bar 0.16.

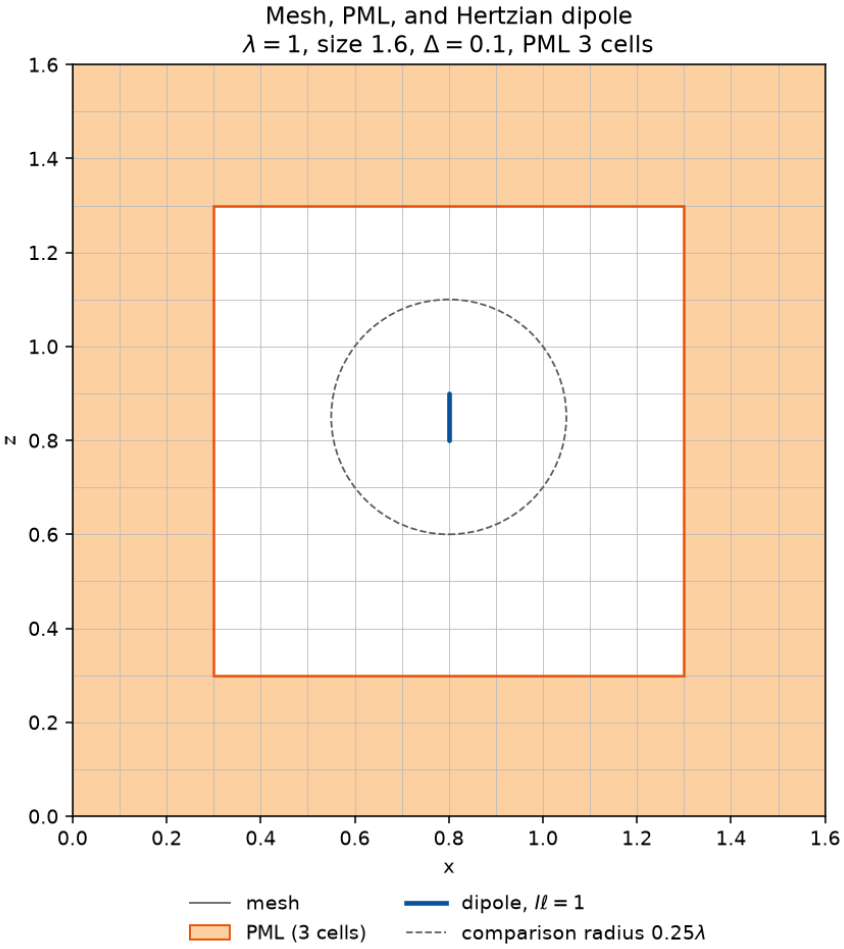

Figure 15: Default Hertzian-dipole domain. The orange frame is the PML band used by the solve. The blue segment is the z-directed element. The dashed circle is the 0.25λ radius inside which the field comparison is omitted.

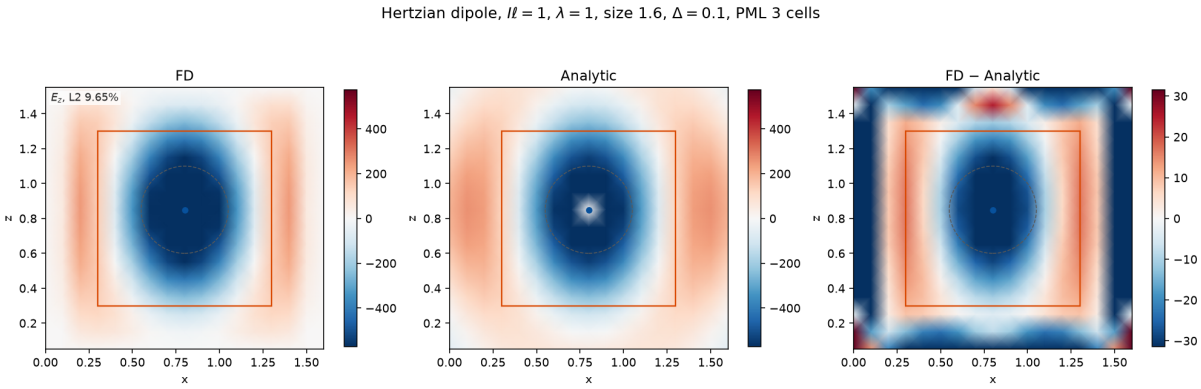

Figure 16: Hertzian dipole of moment $I\ell = 1$ at $\lambda = 1$, on a cube of side 1.6 at $\Delta = 0.1$, with three PML cells on each face. The panels are $\mathrm{Re}\, E_z$ on the xz plane through the element. The annotated L^2 is the electric-sample error outside the PML and at least 0.25λ from the element.

5.9 Gaussian beam

```
python examples/fdfd/2d/gaussian_beam.py
```

The beam propagates in $+\hat{\mathbf{y}}$ with waist radius $w_0 = 1.5$ and amplitude 1. The waist is the center of a PEC rectangle. `-wavelength`, `-width`, `-height`, `-dx`, `-pml`, `-w0`, and `-split` are the free-space wavelength, the two sides, the cell size, the PML depth, the waist radius, and the first total-field E_z row. The defaults are 1, 8, 6, 0.05, 8, 1.5, and 1.5. The impressed current sits on the two E_z rows that touch that split. Below the split the unknown is the scattered field.

The default run writes

```
gaussian_beam_lam1_waist1.5_width8_height6_dx0.05_pml8_split1.5/
```

with `domain.png` and `gaussian_beam.png`. The box is 160×120 cells. Eight PML cells back every face. The figure shows $\mathrm{Re}\, E_z$. On this grid the annotated L^2, outside the PML and at least 0.5λ from the split, is 5.57%. The regression uses the same grid and the bar 0.10. The scattered-field side of that comparison stays under 2×10^{-3}.

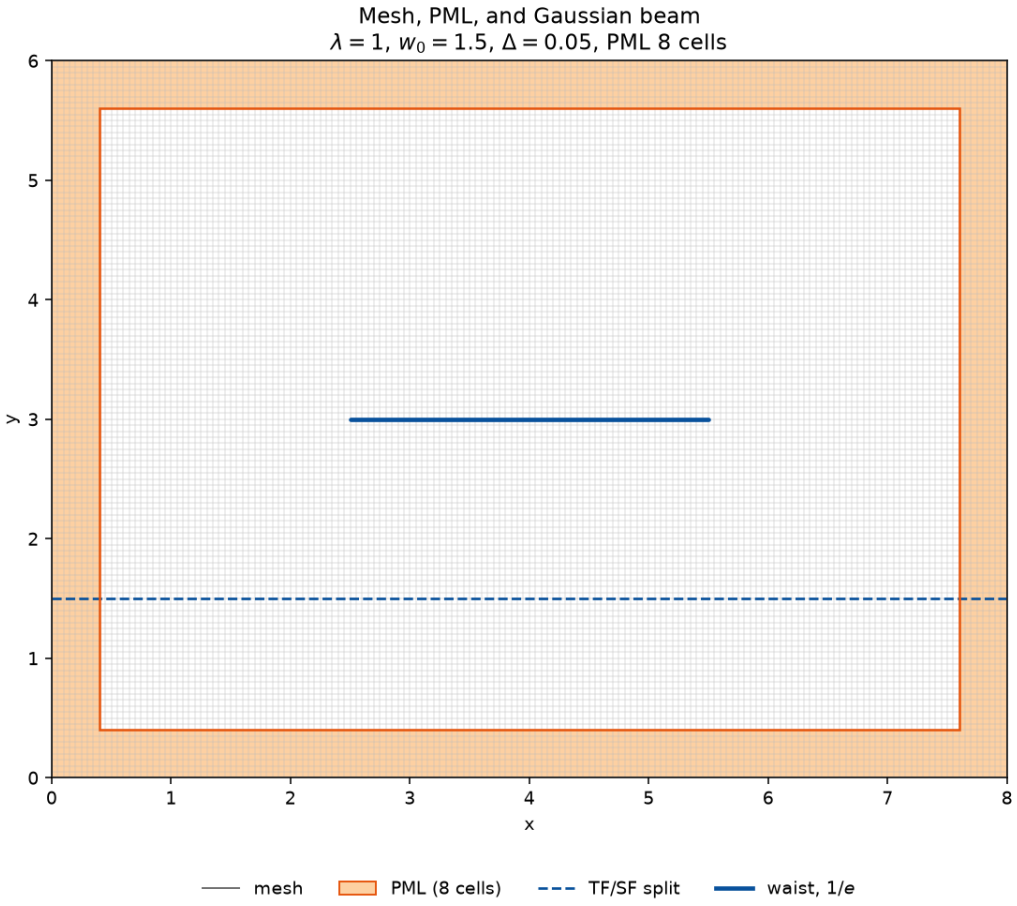

Figure 17: Default Gaussian-beam domain. The orange frame is the PML band used by the solve. The dashed line is the total-field/scattered-field split. The solid segment is the $1/e$ waist.

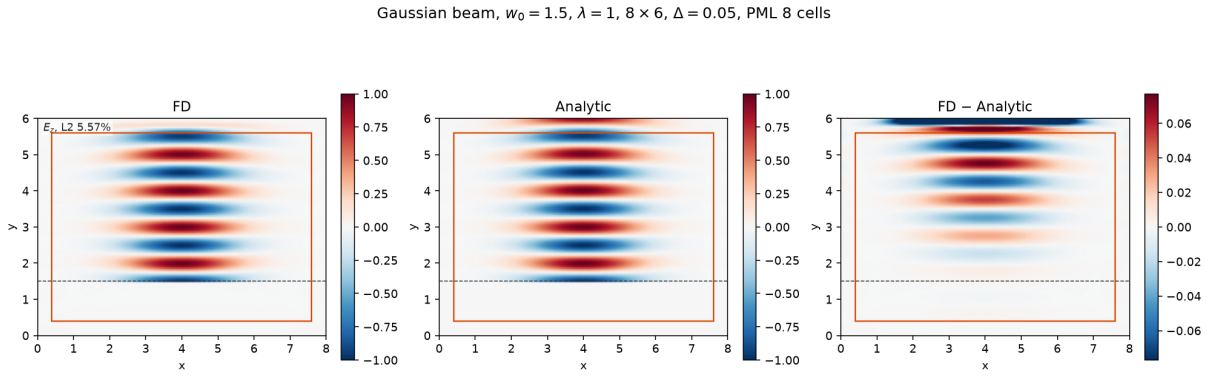

Figure 18: Gaussian beam of waist radius $w_0 = 1.5$ at $\lambda = 1$, on an 8×6 box at $\Delta = 0.05$, with eight PML cells on each face. The panels are $\mathrm{Re}\, E_z$. The dashed line is the split. The annotated L^2 is the electric-sample error on the total-field side, outside the PML and at least 0.5λ from the split.

5.10 Line-current far field

```
python examples/fdfd/2d/farfield_line.py
```

A unit line current sits on the center E_z sample of a PEC square, $J_z = 1/(\Delta x \Delta y)$. A convolutional PML backs every wall. The contour is a square of half-width `-half`, centered on that sample, and it stays in the unstretched interior. E_z is read on the nodes. Tangential H on each edge is the average of the two faces that touch the edge. The pattern at 20 wavelengths is compared with the leading Hankel wave of that current. `-wavelength`, `-side`, `-dx`, `-pml`, and `-half` are the free-space wavelength, the square, the cell size, the PML depth, and the contour half-width. The defaults are 1, 3, 0.05, 8, and 0.8.

The default run writes

```
farfield_line_lam1_side3_dx0.05_pml8_half0.8/
```

with `domain.png` and `farfield_line.png`. The square is 60 cells on a side. Eight PML cells back every face. The pattern figure shows $|E_z|$ and the phase at 72 angles. On this grid the annotated L^2 is 2.62%. The regression uses the same grid and the bar 0.05. Halving the spacing, with the PML deepened to 16 cells on the same physical band, stays under 0.02 and drops the error by more than half.

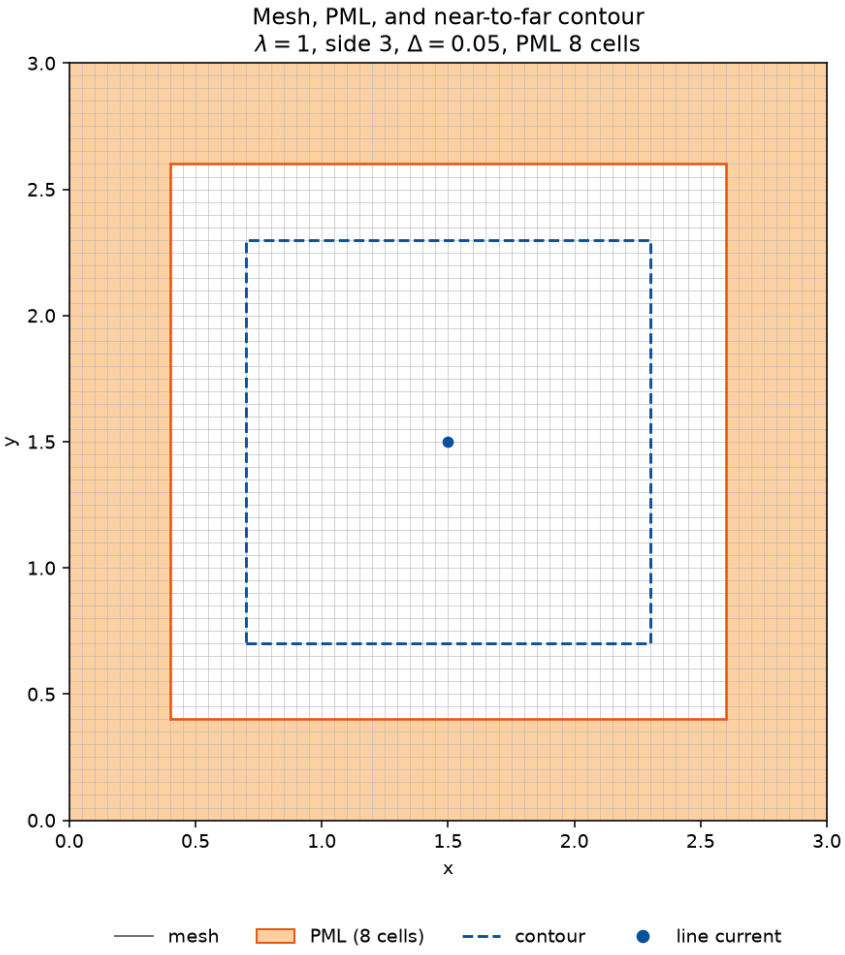

Figure 19: Default line-current far-field domain. The orange frame is the PML band used by the solve. The dashed square is the near-to-far contour. The blue dot is the line current.

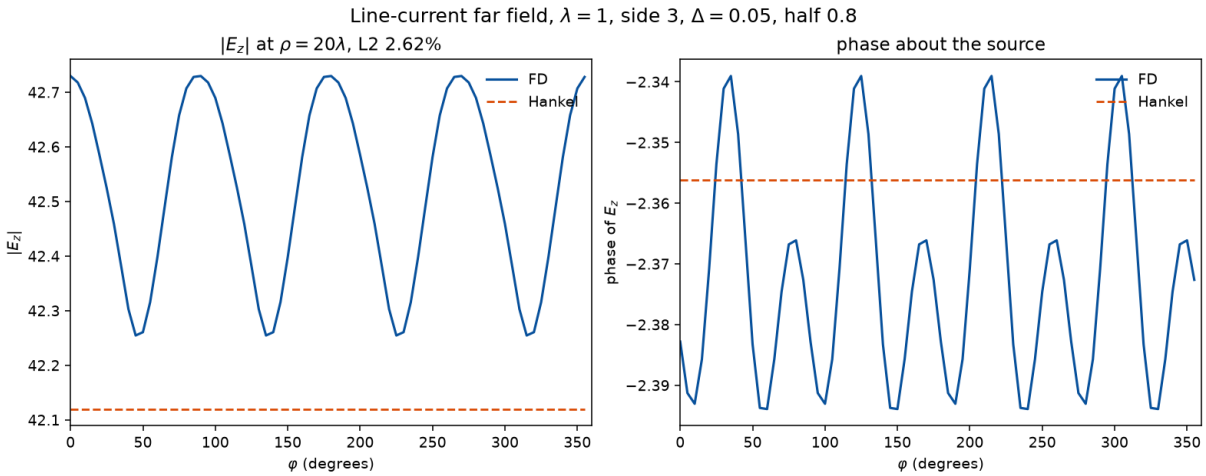

Figure 20: Line-current far field at $\lambda = 1$, on a square of side 3 at $\Delta = 0.05$, with eight PML cells on each face and a contour of half-width 0.8. The panels are $|E_z|$ and the phase at $\rho = 20\lambda$. The annotated L^2 is the complex error against the leading Hankel wave.

5.11 Resistive sheet

```
python examples/fdfd/2d/resistive_sheet.py
```

`-zs` is the surface impedance in ohms per square. The default is η_0. `-wavelength`, `-y`, `-length`, `-height`, and `-dx` are the free-space wavelength, the sheet coordinate, the two sides, and the cell size. Both sides must be a positive integer number of cells, and the sheet must lie in $(0, \texttt{height})$.

The default run writes

```
resistive_sheet_zs376.730313462_lam46_y40_length8_height80_dx1/
```

with `domain.png` and `resistive_sheet.png`. The box is 8×80 cells at $\Delta = 1$. The sheet sits at $y = 40$, on an electric node, with $Z_s = \eta_0$ and $\lambda = 46$. The impressed current is zero. The wall carries the analytic trace. TMz prescribes $E_z(y)$ and TEz prescribes $E_x(y)$ with $E_y = 0$. Figure 1 is the Ampere cell used for this run: the sheet lies on electric row j. The field figure shows $\mathrm{Re}\, E_z$ and $\mathrm{Re}\, E_x$, with propagation along the horizontal axis. On this grid the annotated errors are 0.02% (TMz) and 0.42% (TEz). The regression uses the same grid and the bars 6×10^{-4} and 1.2×10^{-2}.

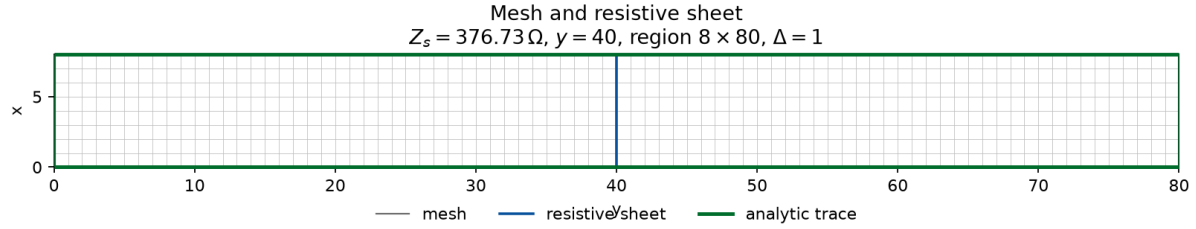

Figure 21: Default resistive-sheet domain. The blue line is the sheet. The green frame is the analytic trace on all four walls.

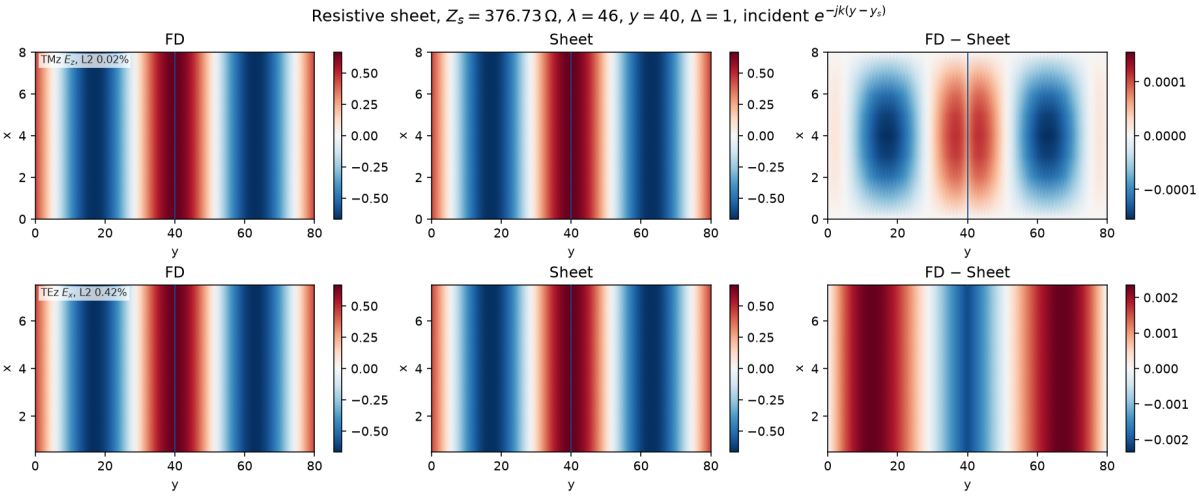

Figure 22: Resistive sheet at $Z_s = \eta_0$, $\lambda = 46$, $y = 40$, on an 8×80 box at $\Delta = 1$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The annotated L^2 is the packed free electric vector.

5.12 Symmetric sheet

```
python examples/fdfd/2d/symmetric_sheet.py
```

`-chi-ee` and `-chi-mm` are the tangential susceptibilities. The defaults are 0.5 and 0.25. `-wavelength`, `-y`, `-length`, `-height`, and `-dx` are the free-space wavelength, the magnetic-face coordinate, the two sides, and the cell size. Both sides must be a positive integer number of cells, and the sheet must lie on one magnetic face inside $(0, \texttt{height})$.

The default run writes

```
symmetric_sheet_chiee0.5_chimm0.25_lam46_y40.5_length8_height80_dx1/
```

with `domain.png` and `symmetric_sheet.png`. The box is 8×80 cells at $\Delta = 1$. The sheet sits at $y = 40.5$, halfway between electric rows, with $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, and $\lambda = 46$. Figure 2 is that face. The impressed current is zero. The wall carries the analytic trace. TMz prescribes $E_z(y)$ and TEz prescribes $E_x(y)$ with $E_y = 0$. The field figure shows $\mathrm{Re}\, E_z$ and $\mathrm{Re}\, E_x$, with propagation along the horizontal axis. On this grid the annotated errors are 1.39% (TMz) and 6.39% (TEz). The regression uses the same grid and the bars 4.2×10^{-2} and 1.9×10^{-1}. A cell of $\Delta = 1/3$ on the same sheet drops both errors by more than half.

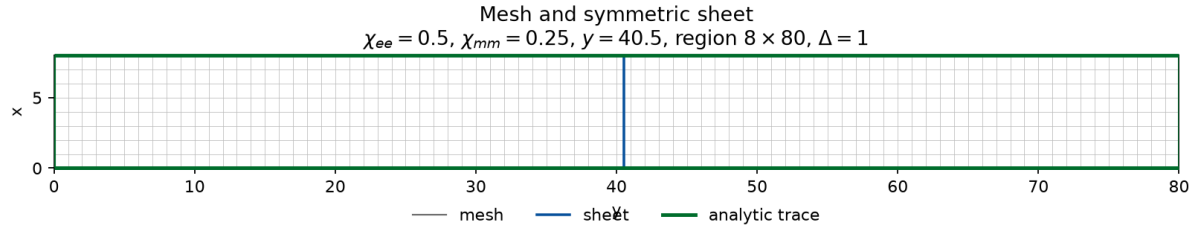

Figure 23: Default symmetric-sheet domain. The blue line is the sheet on a magnetic face. The green frame is the analytic trace on all four walls.

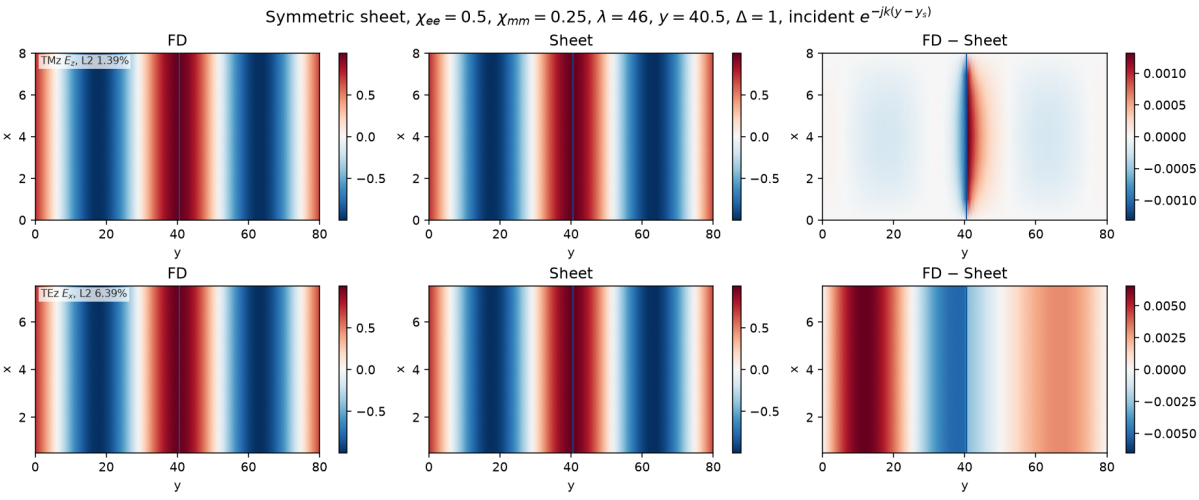

Figure 24: Symmetric sheet at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\lambda = 46$, $y = 40.5$, on an 8×80 box at $\Delta = 1$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The annotated L^2 is the packed free electric vector.

5.13 Finite absorbing sheet

```
python examples/fdfd/2d/finite_sheet.py
```

`-sheet` is the sheet length in wavelengths. The default is 10. `-theta` is the incidence angle in degrees, measured from the sheet normal, and the default is 0. `-wavelength`, `-points`, and

`-pml` are the free-space wavelength, the cells per wavelength, and the PML depth on each side. The susceptibilities are $\chi_{ee} = \chi_{mm} = -2j/k$, so a uniform sheet of this normal has $R = T = 0$ at $\theta = 0$. Passing `-chi-ee`, `-chi-mm`, and `-chi-nn` together replaces that absorber. `-chi-em` and `-chi-me` default to zero. The absorber rejects a nonzero value. Section 5.18 sets them with the three real susceptibilities. Section 5.16 is that run at 30°. The domain leaves two wavelengths of vacuum between the sheet and the PML. The sheet lies on the magnetic face half a cell below mid-height. The local jump is the normal-incidence cell of Figure 2. The angle enters only through the incident phase $e^{-jk(\sin\theta\,(x-x_c)+\cos\theta\,(y-y_s))}$, with phase zero at the sheet center.

The default run writes

`finite_sheet_lam1_points20_theta0_pml12_sheet10/`

with `domain.png` and `finite_sheet.png`. The sheet is 10λ long at 20 cells per wavelength, with 12 PML cells on each side and $\theta = 0$. The unknown is the scattered field. The incident electric amplitude is 1. The field figure shows $\mathrm{Re}\,E_z$ (TMz) and $\mathrm{Re}\,E_x$ (TEz). On this grid the annotated means, over the central 6λ of the sheet, are a lit-side $|E|$ of 1.02 and a shadow $|E|$ of 0.03 (TMz) and 0.02 (TEz).

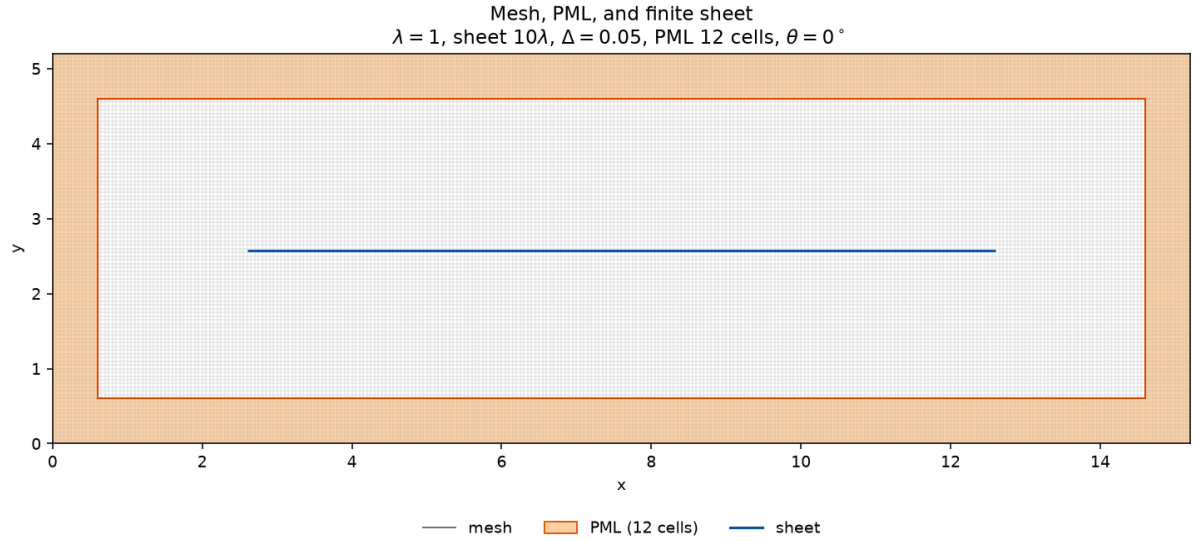

Figure 25: Default finite-sheet domain. The blue segment is the 10λ sheet on a magnetic face. The orange frame is the PML band on all four sides. The wall behind that band is PEC.

The same absorber at 30° is

```
python examples/fdfd/2d/finite_sheet.py --theta 30
```

and writes

`finite_sheet_lam1_points20_theta30_pml12_sheet10/`.

The susceptibilities are still the normal-incidence pair $\chi_{ee} = \chi_{mm} = -2j/k$. On this grid the annotated means are a lit-side $|E|$ of 1.02 and a shadow $|E|$ of 0.04 (TMz), and 0.86 and 0.01 (TEz). The lit TEz value follows $\cos 30°$ because the plotted component is E_x.

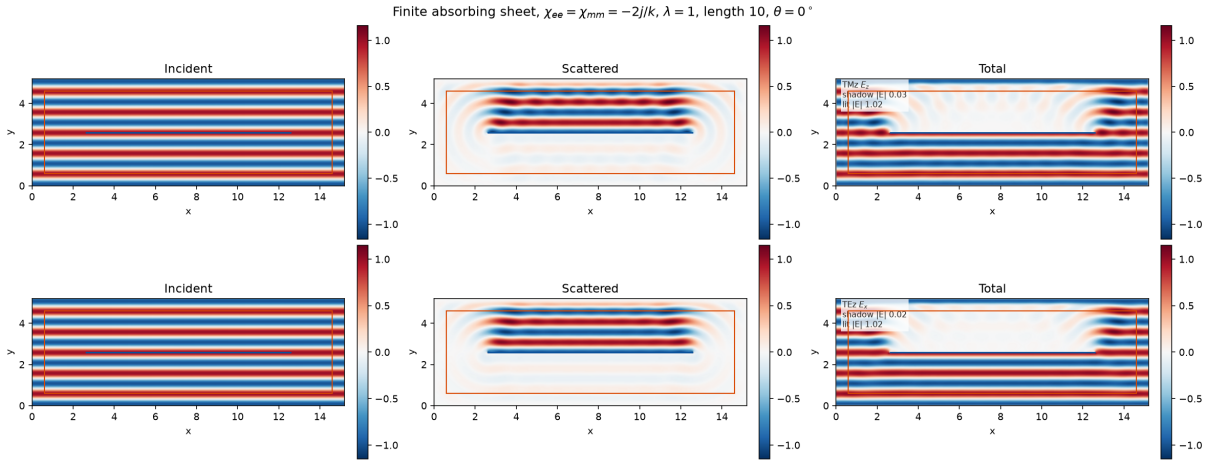

Figure 26: Finite absorbing sheet, $\chi_{ee} = \chi_{mm} = -2j/k$, $\lambda = 1$, length 10λ, $\theta = 0$, at 20 cells per wavelength with 12 PML cells on each side. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the incident wave, the scattered field, and the total field. The annotated means are the lit side and the shadow over the central 6λ of the sheet.

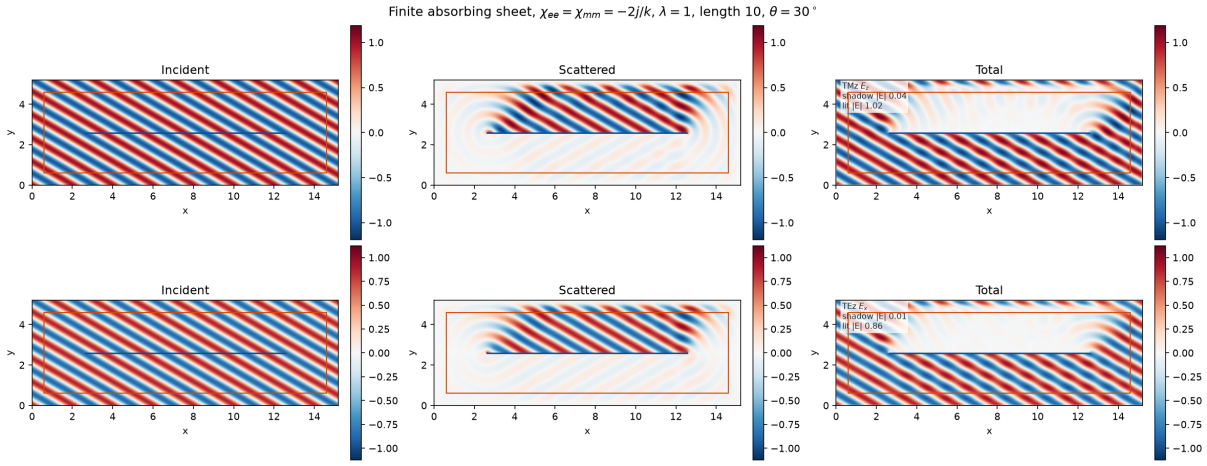

Figure 27: Finite absorbing sheet at $\theta = 30°$, with $\chi_{ee} = \chi_{mm} = -2j/k$, $\lambda = 1$, and length 10λ. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The annotated means are the lit side and the shadow over the central 6λ of the sheet.

5.14 Periodic oblique sheet

```
python examples/fdfd/2d/periodic_sheet.py
```

`-chi-ee`, `-chi-mm`, and `-chi-nn` are the tangential electric susceptibility, the tangential magnetic susceptibility, and the normal magnetic susceptibility. The defaults are 0.5, 0.25, and 0.4. `-theta` is the incidence angle in degrees, measured from the sheet normal, and the default is 30. `-wavelength`, `-points`, and `-height` are the free-space wavelength, the cells per wavelength, and the domain height in wavelengths. The defaults are 1, 20, and 4. `-chi-em` and `-chi-me` are the magneto-electric strengths. They default to zero, and a zero stays out of the folder name. Section 5.17 sets both to 0.2.

The sheet spans the whole magnetic face, half a cell below mid-height. The width is one transverse period, $\lambda/\sin\theta$, so the phase of an oblique wave closes under a periodic identification of phase one. `Boundary.PERIODIC_X` is that identification in x and a PEC wall in y. The y ends carry the analytic total field. χ^{nn}_{mm} adds $\chi^{nn}_{mm}\partial_x H_{y,\mathrm{av}}$ on the TMz jump. The derivative uses the bulk H_y, recovered from E_z. TEz keeps the tangential pair, so its reflection and transmission follow χ_{ee} and χ_{mm}.

The default run writes

```
periodic_sheet_lam1_theta30_points20_height4_chiee0.5_chimm0.25_chinn0.4/
```

with `domain.png` and `periodic_sheet.png`. At 30° the cell is 2λ wide and 4λ tall, a 40×80 grid at $\Delta = 0.05$. The field figure shows $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are FD, Analytic, and FD−Analytic. On this grid the annotated L^2 errors are 9.63% (TMz) and 8.91% (TEz). The regression uses the same susceptibilities at 10 cells per wavelength, with the bars 0.24 and 0.20.

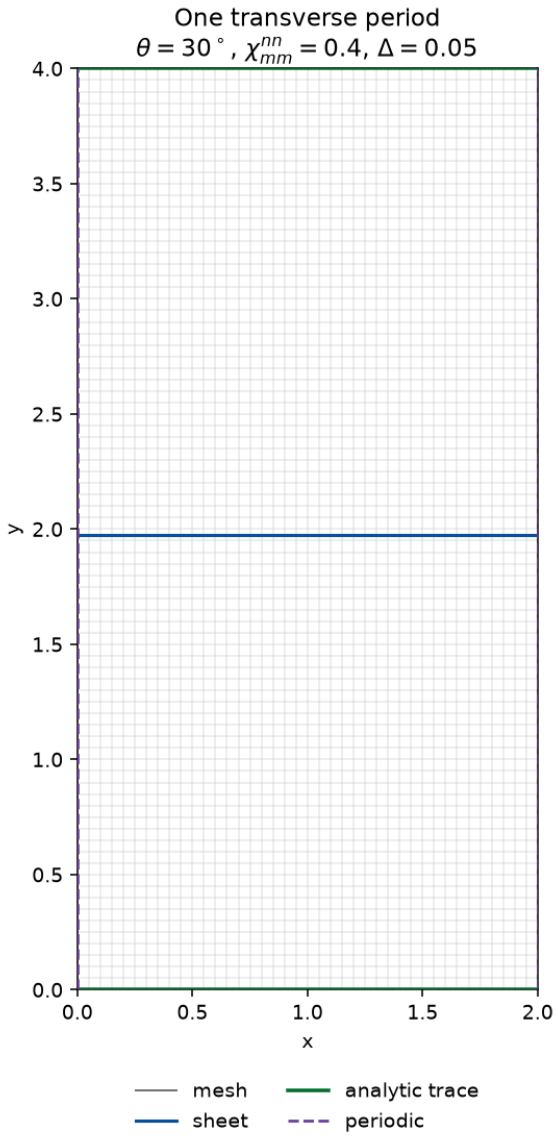

Figure 28: Default periodic-sheet domain. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed purple lines mark the periodic identification of the x ends. The width is one transverse period at 30°.

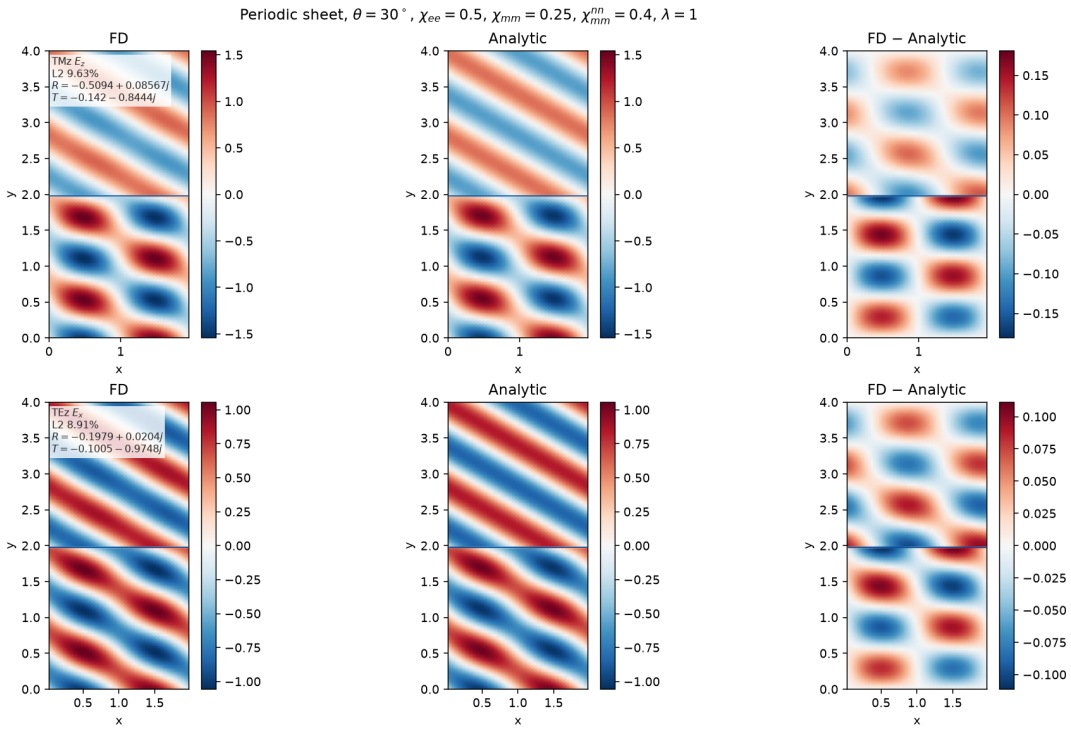

Figure 29: Periodic oblique sheet at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi_{mm}^{nn} = 0.4$, $\lambda = 1$, $\theta = 30°$, and a height of 4λ, on a 40×80 grid at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated L^2 is the packed free electric vector.

5.15 Bloch unit cell

```
python examples/fdfd/2d/bloch_sheet.py
```

`-chi-ee`, `-chi-mm`, and `-chi-nn` are the tangential electric susceptibility, the tangential magnetic susceptibility, and the normal magnetic susceptibility. The defaults are 0.5, 0.25, and 0.4, the same sheet as Sec. 5.14. `-theta` is the incidence angle in degrees, measured from the sheet normal, and the default is 30. `-wavelength`, `-points`, and `-height` default to 1, 20, and 4. `-width` is the cell width in wavelengths, and the default is 0.8. `-chi-em` and `-chi-me` default to zero and stay out of the folder name at that default.

The sheet spans the magnetic face. The width is 0.8λ, which is 0.4 of the transverse period at 30°, so a phase of one does not close the oblique wave. `Boundary.PERIODIC_X` identifies the x ends, and the grid sets $k_{B,x} = k \sin\theta$. The forward wrap then carries $e^{-j\phi}$ with $\phi = k_{B,x} a$. On this grid $\phi = 0.8\pi$ and that factor is $-0.809 - 0.588j$. The backward wrap carries the conjugate. The y ends carry the analytic total field. The reflection and transmission are those of the uniform sheet in Sec. 5.16.

The default run writes

```
bloch_sheet_lam1_theta30_width0.8_points20_height4_chiee0.5_chimm0.25_chinn0.4/
```

with `domain.png` and `bloch_sheet.png`. The cell is 0.8λ wide and 4λ tall, a 16×80 grid at $\Delta = 0.05$. The field figure shows $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are FD, Analytic, and FD−Analytic. On this grid the annotated L^2 errors are 9.63% (TMz) and 8.91% (TEz). The regression uses the same susceptibilities at 10 cells per wavelength, with the bars 0.22 and 0.18. Four times that resolution drops both errors by more than half. The same cell with phase one stays above 0.5 in both polarizations. The orders $m = \pm 1$ of the demodulated field, on an electric row a quarter of the way up the cell, stay under 10^{-8} of the specular amplitude.

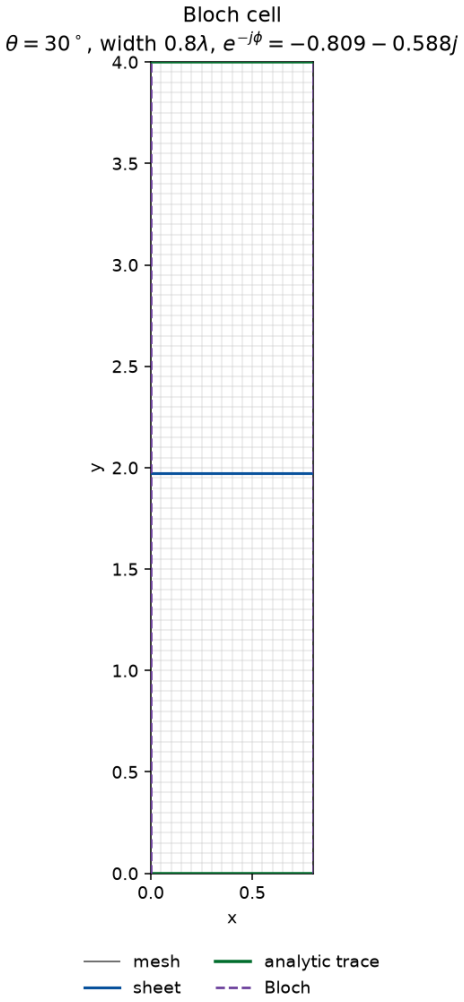

Figure 30: Default Bloch-cell domain. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed purple lines are the Bloch identification of the x ends. The width is 0.8 wavelengths, and the title prints the forward factor.

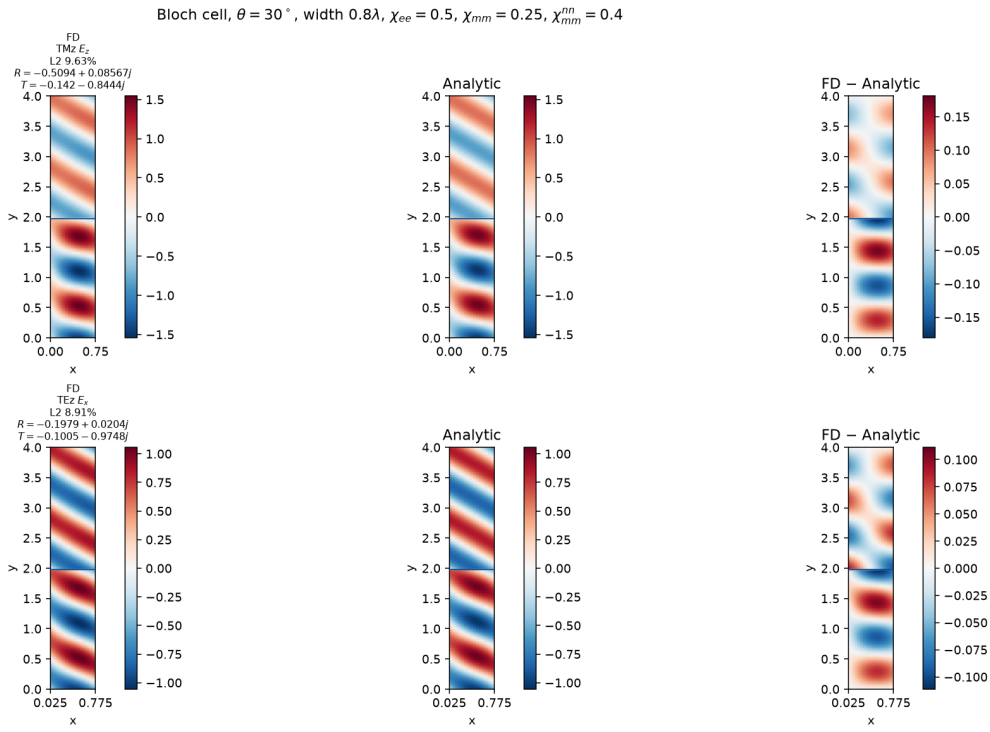

Figure 31: Bloch cell at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi_{mm}^{nn} = 0.4$, $\lambda = 1$, $\theta = 30°$, width 0.8λ, and height 4λ, on a 16×80 grid at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated L^2 is the packed free electric vector, and the FD title prints the analytic reflection and transmission.

5.16 Finite oblique sheet

```
python examples/fdfd/2d/finite_sheet.py \
    --chi-ee 0.5 --chi-mm 0.25 --chi-nn 0.4 --theta 30
```

The geometry is the 10λ sheet of Sec. 5.13: 20 cells per wavelength, 12 PML cells on each side, and two wavelengths of vacuum between the sheet and the band. The susceptibilities are the ones in Sec. 5.14, and the incidence is 30°. The unknown is the scattered field. The angle enters through the incident phase. On the TMz jump, χ^{nn}_{mm} also adds $\chi^{nn}_{mm}\partial_x H_{y,\mathrm{av}}$. TEz keeps the tangential pair. A uniform sheet with these values has $R = -0.5094 + 0.08567j$ and $T = -0.1420 - 0.8444j$ in TMz, and $R = -0.1979 + 0.02040j$ and $T = -0.1005 - 0.9748j$ in TEz.

The run writes

```
finite_sheet_lam1_points20_theta30_pml12_sheet10_chiee0.5_chimm0.25_chinn0.4/
```

with `domain.png` and `finite_sheet.png`. The field figure shows Re E_z (TMz) and Re E_x (TEz). The columns are the incident wave, the scattered field, and the total field. On this grid the annotated means, over the central 6λ of the sheet, are a lit-side $|E|$ of 1.15 and a shadow $|E|$ of 0.85 (TMz), and 0.90 and 0.84 (TEz). A uniform sheet transmits $|E_z| = 0.86$ and $|E_x| = 0.85$. The shadow of the finite sheet sits on those values, and the ends diffract.

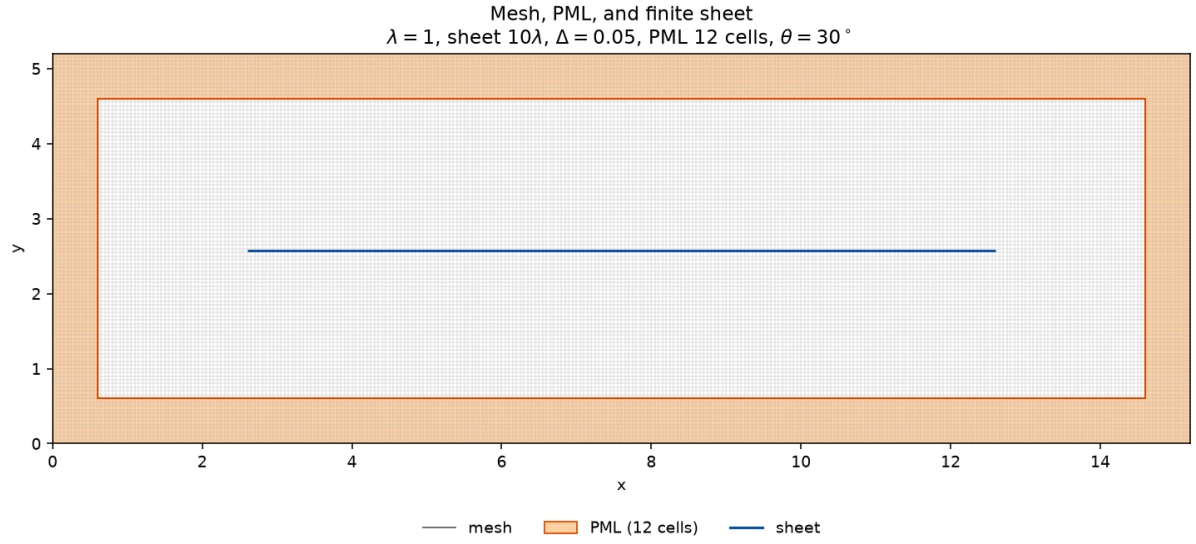

Figure 32: Finite oblique-sheet domain. The blue segment is the 10λ sheet on a magnetic face. The orange frame is the PML band on all four sides. The incidence is 30° and the susceptibilities are $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi^{nn}_{mm} = 0.4$.

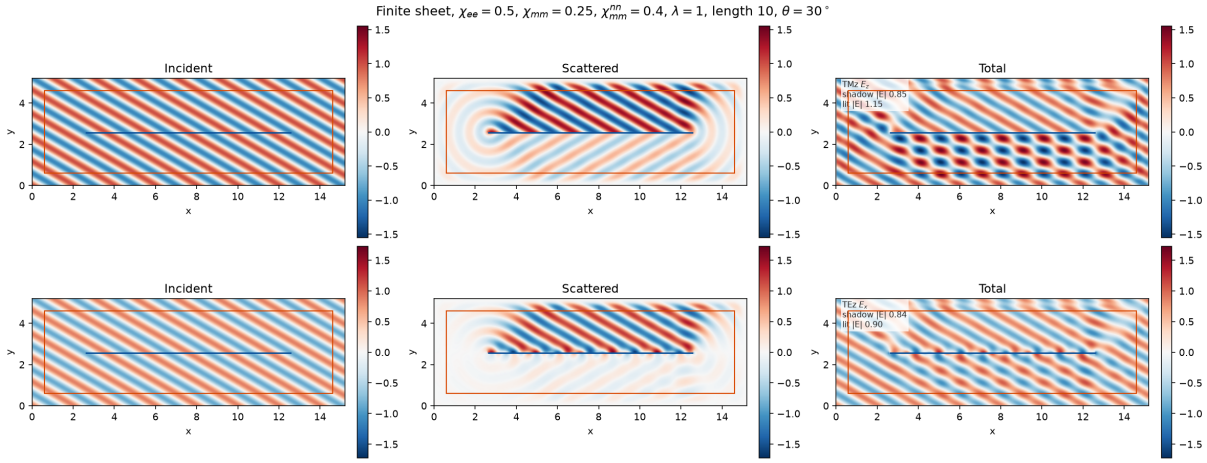

Figure 33: Finite sheet at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi_{mm}^{nn} = 0.4$, $\lambda = 1$, length 10λ, $\theta = 30°$, at 20 cells per wavelength with 12 PML cells on each side. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the incident wave, the scattered field, and the total field. The annotated means are the lit side and the shadow over the central 6λ of the sheet.

5.17 Periodic magneto-electric sheet

```
python examples/fdfd/2d/periodic_sheet.py --chi-em 0.2 --chi-me 0.2
```

The geometry and the tangential susceptibilities are those of Sec. 5.14. $\chi_{em} = \chi_{me} = 0.2$ is the reciprocal magneto-electric pair, and both polarizations carry it. χ^{nn}_{mm} remains on the TMz jump. A uniform sheet with these values has $R = -0.07672 + 0.02351j$ and $T = -0.2819 - 0.9201j$ in TMz, and $R = 0.329 - 0.09119j$ and $T = -0.2984 - 1.077j$ in TEz. The reflective lock $\chi_{em} = \chi_{me} = -2j/k$, with the other susceptibilities zero, gives $R = 1$ and $T = 0$ at normal incidence.

The run writes

```
periodic_sheet_lam1_theta30_points20_height4_chiee0.5_chimm0.25_chinn0.4_chiem0.
                                    2_chime0.2/
```

with `domain.png` and `periodic_sheet.png`. The cell is again 2λ wide and 4λ tall, a 40×80 grid at $\Delta = 0.05$. The field figure shows $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are FD, Analytic, and FD$-$Analytic. On this grid the annotated L^2 errors are 8.38% (TMz) and 7.62% (TEz). The regression uses the same susceptibilities at 10 cells per wavelength, with the bars 0.24 and 0.21.

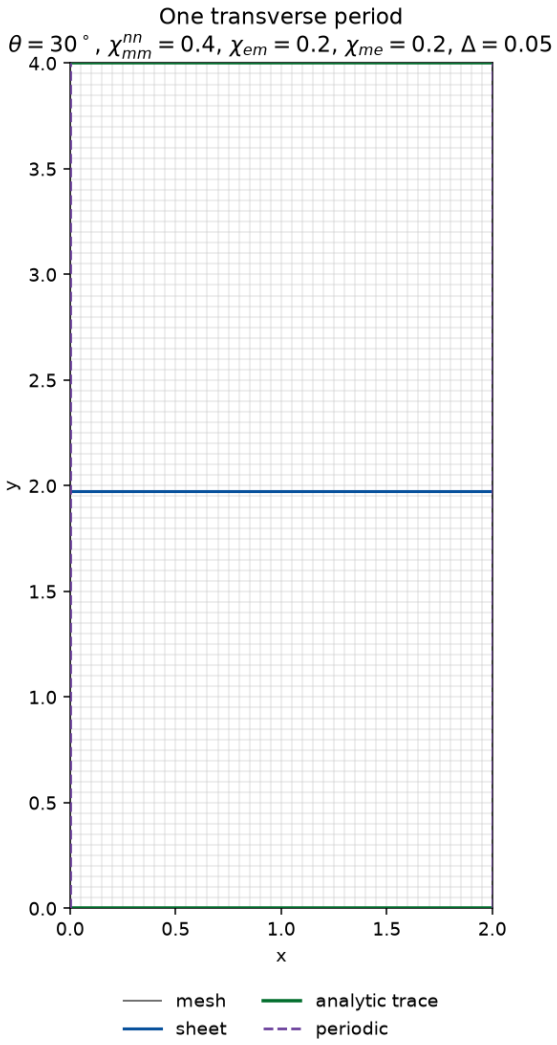

Figure 34: Periodic magneto-electric domain. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed purple lines mark the periodic identification of the x ends. The susceptibilities add $\chi_{em} = \chi_{me} = 0.2$ to the tangential and normal values of Figure 28.

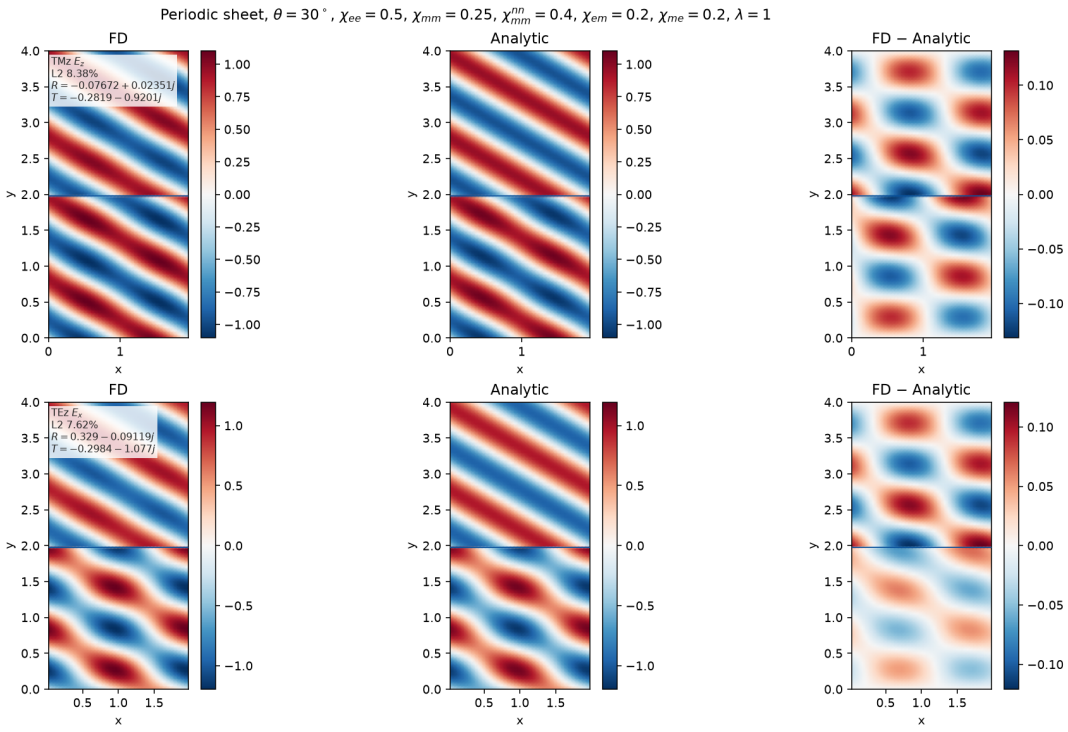

Figure 35: Periodic oblique sheet at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi_{mm}^{nn} = 0.4$, $\chi_{em} = \chi_{me} = 0.2$, $\lambda = 1$, $\theta = 30°$, and a height of 4λ, on a 40×80 grid at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated L^2 is the packed free electric vector.

5.18 Finite magneto-electric sheet

```
python examples/fdfd/2d/finite_sheet.py \
    --chi-ee 0.5 --chi-mm 0.25 --chi-nn 0.4 \
    --chi-em 0.2 --chi-me 0.2 --theta 30
```

The geometry is the 10λ sheet of Sec. 5.13: 20 cells per wavelength, 12 PML cells on each side, and two wavelengths of vacuum between the sheet and the band. The susceptibilities and the incidence are those of Sec. 5.17. The unknown is the scattered field. A uniform sheet transmits $|E_z| = 0.96$ and $|E_x| = 0.97$.

The run writes

```
finite_sheet_lam1_points20_theta30_pml12_sheet10_chiee0.5_chimm0.25_chinn0.4_
                                chiem0.2_chime0.2/
```

with `domain.png` and `finite_sheet.png`. The field figure shows Re E_z (TMz) and Re E_x (TEz). The columns are the incident wave, the scattered field, and the total field. On this grid the annotated means, over the central 6λ of the sheet, are a lit-side $|E|$ of 1.01 and a shadow $|E|$ of 0.97 (TMz), and 0.85 and 0.96 (TEz). The shadow sits on the uniform transmission, and the ends diffract.

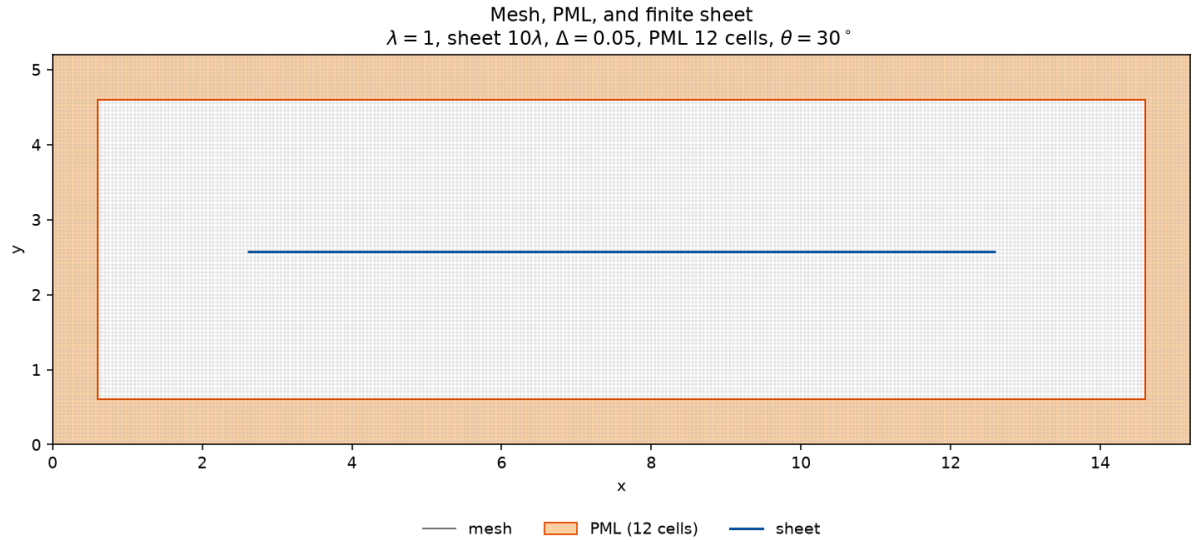

Figure 36: Finite magneto-electric domain. The blue segment is the 10λ sheet on a magnetic face. The orange frame is the PML band on all four sides. The incidence is 30° and the susceptibilities add $\chi_{em} = \chi_{me} = 0.2$ to the values of Figure 32.

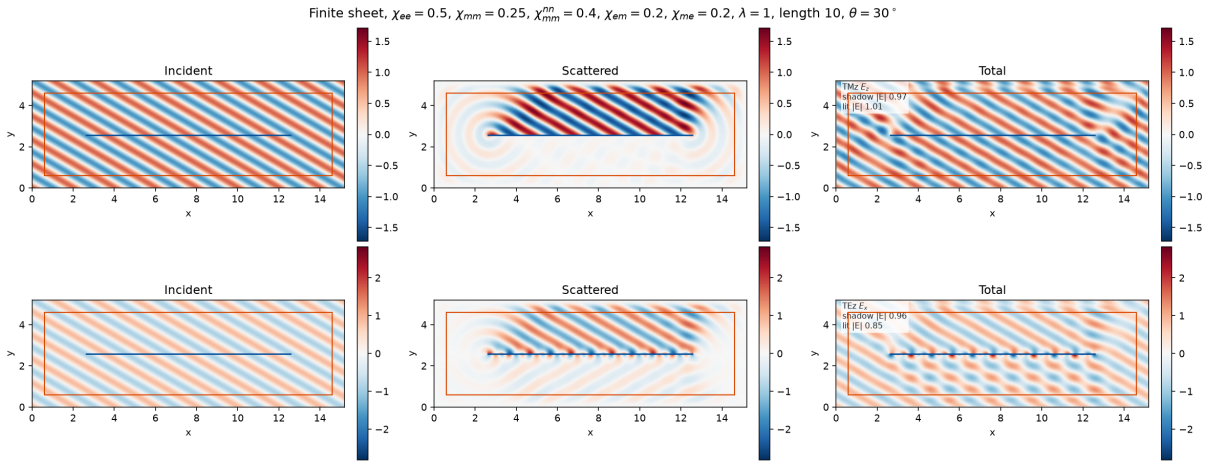

Figure 37: Finite sheet at $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi^{nn}_{mm} = 0.4$, $\chi_{em} = \chi_{me} = 0.2$, $\lambda = 1$, length 10λ, $\theta = 30°$, at 20 cells per wavelength with 12 PML cells on each side. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the incident wave, the scattered field, and the total field. The annotated means are the lit side and the shadow over the central 6λ of the sheet.

5.19 Two-piece sheet

```
python examples/fdfd/2d/piecewise_sheet.py
```

The sheet is 10λ long at 20 cells per wavelength, with 12 PML cells on each side and two wavelengths of vacuum between the sheet and the band. `-split` is the left piece, 5λ by default. That piece is the absorber $\chi_{ee} = \chi_{mm} = -2j/k$. The right piece is $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$. The incidence is normal, so a uniform sheet of either piece has the same coefficients in both polarizations. The absorber has $R = T = 0$. The right-hand pair has $|R| = 0.33$ and $|T| = 0.94$. The unknown is the scattered field. Each piece is compared with the uniform sheet of its own susceptibilities, on a window one wavelength inside the ends and the junction. The absorber shadow is a mean $|E|$, because that transmission is zero. The other three windows are a relative L^2.

The run writes

```
piecewise_sheet_lam1_points20_theta0_pml12_sheet10_split5/
```

with `domain.png` and `piecewise_sheet.png`. The field figure shows $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the incident wave, the scattered field, and the total field. On this grid the annotated windows are a left lit L^2 of 0.09 and a left shadow $|E|$ of 0.09, then a right lit L^2 of 0.08 and a right shadow L^2 of 0.20 (TMz), and 0.08, 0.04, 0.07, and 0.20 (TEz). The shadow of the absorber is dark. The right-hand piece carries the transmitted wave, and the junction and the ends diffract.

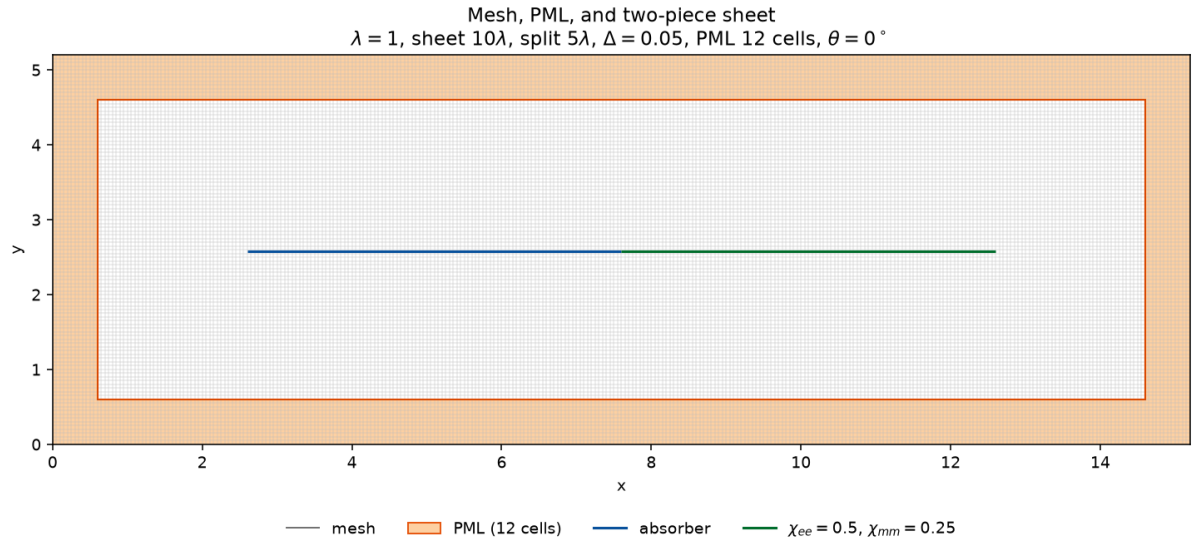

Figure 38: Two-piece domain. The blue segment is the absorbing half and the green segment is $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$. The orange frame is the PML band on all four sides. The sheet is 10λ long and the split is at 5λ.

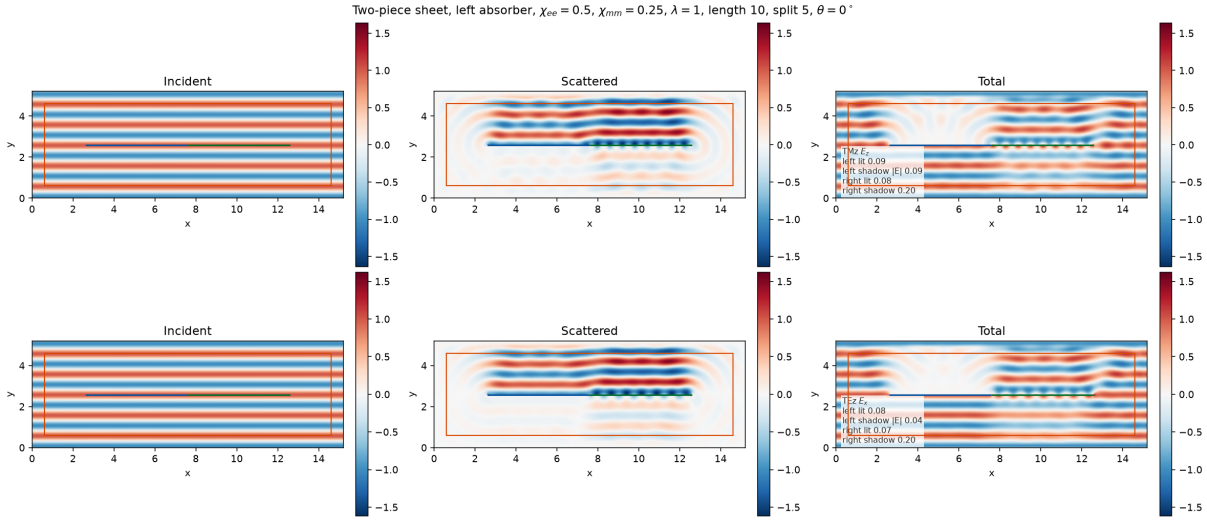

Figure 39: Two-piece sheet, $\lambda = 1$, length 10λ, split 5λ, $\theta = 0°$, at 20 cells per wavelength with 12 PML cells on each side. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the incident wave, the scattered field, and the total field. The annotation is the interior window of each piece: lit-side relative L^2, absorber-shadow mean $|E|$, then the right-hand lit and shadow relative L^2.

5.20 Dispersive periodic sheet

```
python examples/fdfd/2d/dispersive_sheet.py
```

The cell is one transverse period wide and 4λ tall, at 20 cells per wavelength and $\theta = 30°$. Phase one on the periodic x wrap closes the oblique wave. The y ends carry the analytic trace. $\chi_{ee0} = 0.5$, $\chi_{ee2} = 0.01$, and $\chi_{mm} = 0.25$. The operator is the normal susceptibility $\chi^{nn}_{mm} = \chi_{ee2}k^2 = 0.395$. At this angle the TMz susceptibility is $\chi_{ee0} + \chi_{ee2}k^2 \sin^2\theta = 0.599$. TEz keeps χ_{ee0}, because the normal term does not enter its jump.

The run writes

```
dispersive_sheet_lam1_theta30_points20_height4_chiee0.5_chimm0.25_chiee20.01/
```

with `domain.png` and `dispersive_sheet.png`. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated packed-electric error is 9.62% (TMz) and 8.91% (TEz).

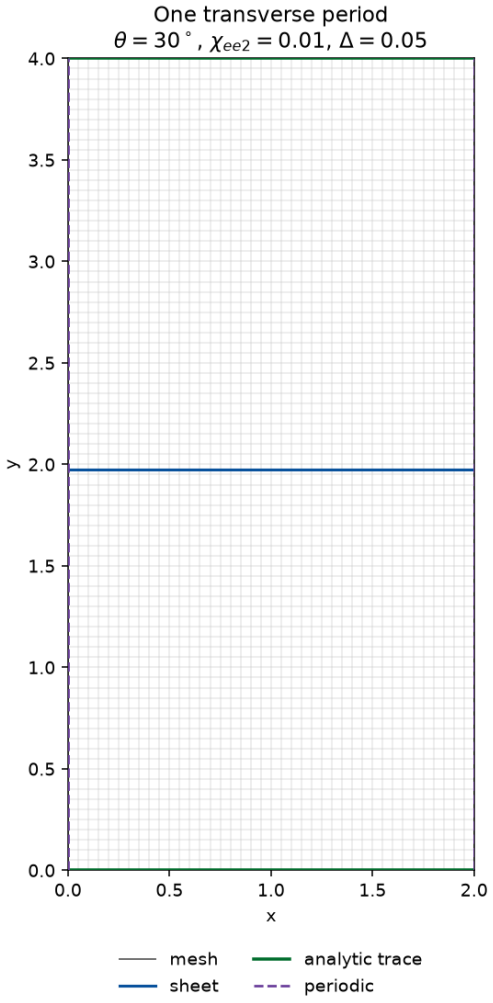

Figure 40: Dispersive periodic cell. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed sides are periodic with phase one. The width is one transverse period at 30°.

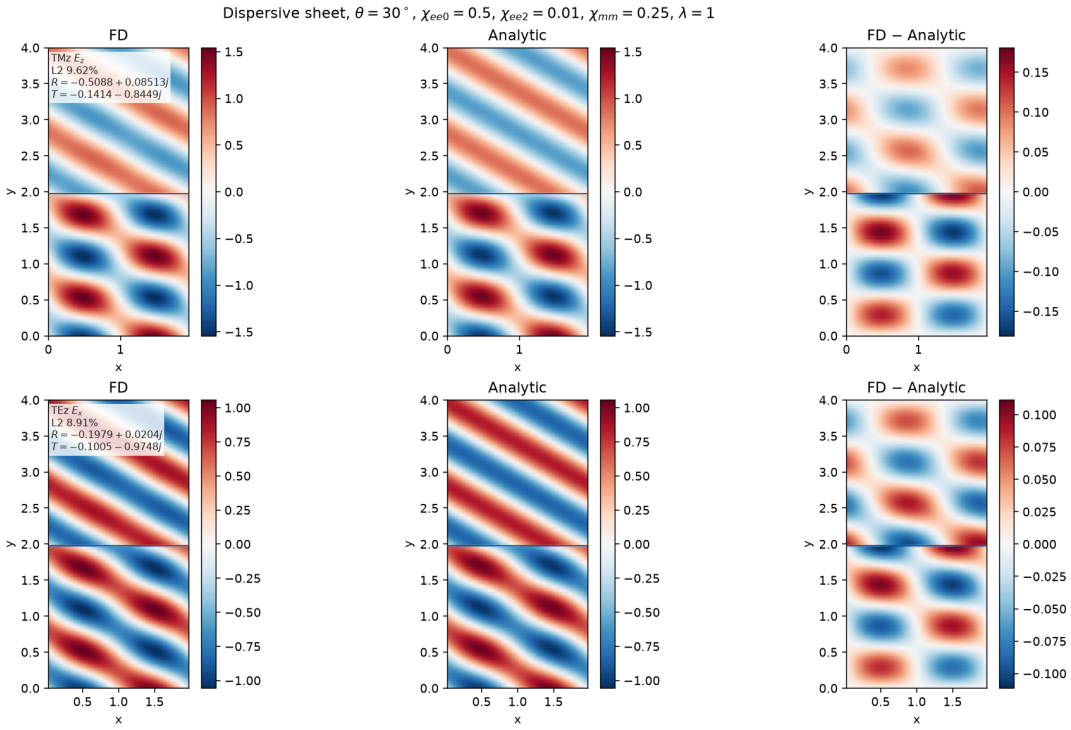

Figure 41: Dispersive sheet at $\chi_{ee0} = 0.5$, $\chi_{ee2} = 0.01$, $\chi_{mm} = 0.25$, $\lambda = 1$, $\theta = 30°$, and a height of 4λ, on a cell one transverse period wide at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. TMz uses $\chi_{ee}(k \sin\theta)$. TEz uses χ_{ee0}. The annotated L^2 is the packed free electric vector.

5.21 Magnetically dispersive periodic sheet

`python examples/fdfd/2d/magnetic_dispersive_sheet.py`

The cell is one transverse period wide and 4λ tall, at 20 cells per wavelength and $\theta = 30°$. Phase one on the periodic x wrap closes the oblique wave. The y ends carry the analytic trace. $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, and $\chi_{mm2} = 0.01$. The operator replaces $\chi_{mm} H$ on the electric jump by $\chi_{mm0} H - \chi_{mm2} \partial_{xx} H$, in both polarizations. At this angle the susceptibility is $\chi_{mm0} + \chi_{mm2} k^2 \sin^2 \theta = 0.349$.

The run writes

`magnetic_dispersive_sheet_lam1_theta30_points20_height4_chiee0.5_chimm0.25_chimm20.01/`

with `domain.png` and `magnetic_dispersive_sheet.png`. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated packed-electric error is 8.35% (TMz) and 8.12% (TEz).

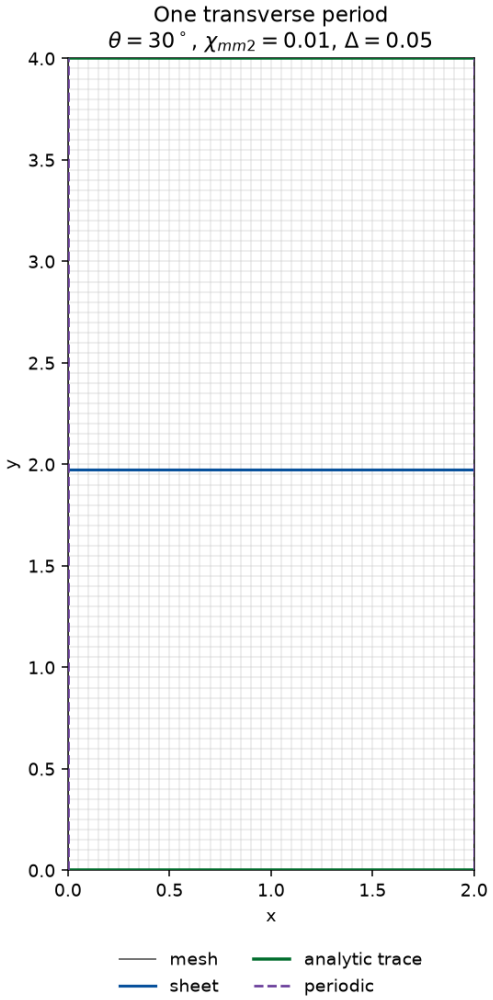

Figure 42: Magnetically dispersive periodic cell. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed sides are periodic with phase one. The width is one transverse period at 30°.

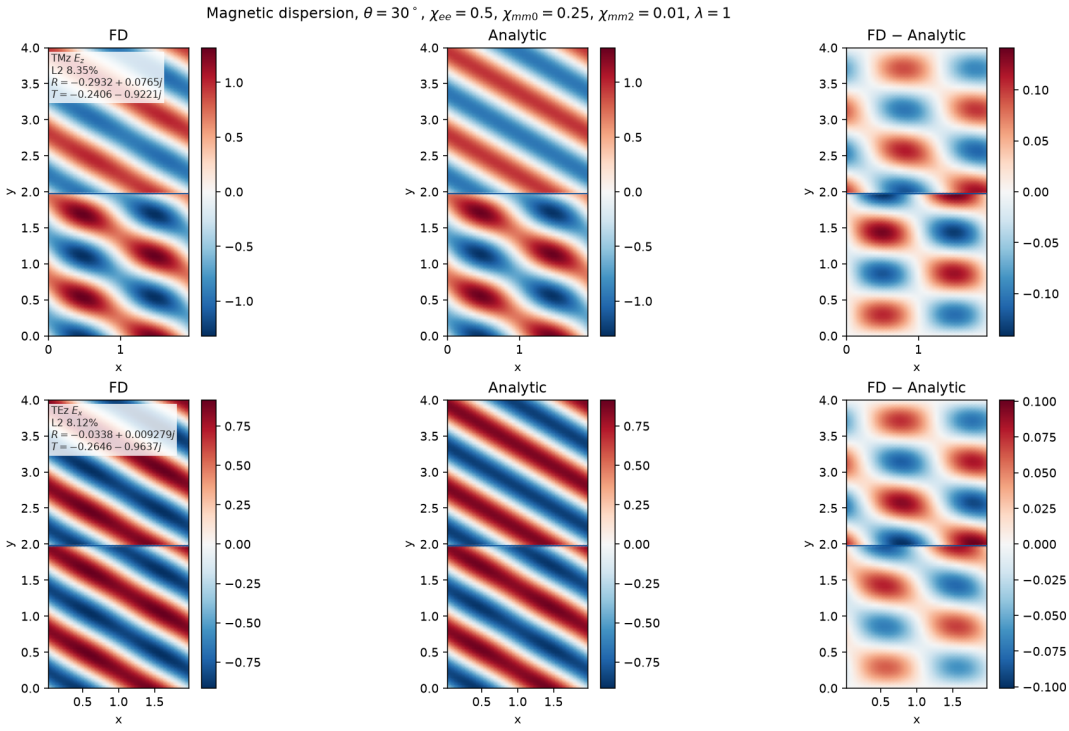

Figure 43: Magnetically dispersive sheet at $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, $\chi_{mm2} = 0.01$, $\lambda = 1$, $\theta = 30°$, and a height of 4λ, on a cell one transverse period wide at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. Both polarizations use $\chi_{mm}(k \sin\theta)$. The annotated L^2 is the packed free electric vector.

5.22 Rational magnetic periodic sheet

```
python examples/fdfd/2d/magnetic_rational_sheet.py
```

The cell is one transverse period wide and 4λ tall, at 20 cells per wavelength and $\theta = 30°$. Phase one on the periodic x wrap closes the oblique wave. The y ends carry the analytic trace. $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, $\chi_{mm2} = 0.04$, and $\xi_{mm2} = 0.04$. One auxiliary per cut replaces $\chi_{mm}H$ on the electric jump, in both polarizations. At this angle the susceptibility is 0.462.

The run writes

```
magnetic_rational_sheet_lam1_theta30_points20_height4_chiee0.5_chimm0.25_
chimm20.04_ximm20.04/
```

with `domain.png` and `magnetic_rational_sheet.png`. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. The annotated packed-electric error is 7.68% (TMz) and 7.80% (TEz).

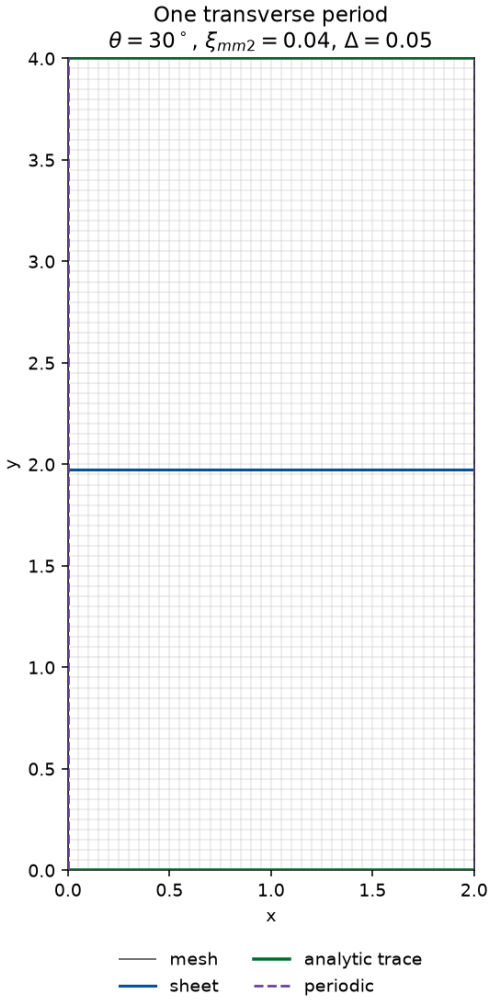

Figure 44: Rational magnetic periodic cell. The blue line is the sheet on a magnetic face. The green lines are the analytic trace on the y ends. The dashed sides are periodic with phase one. The width is one transverse period at 30°.

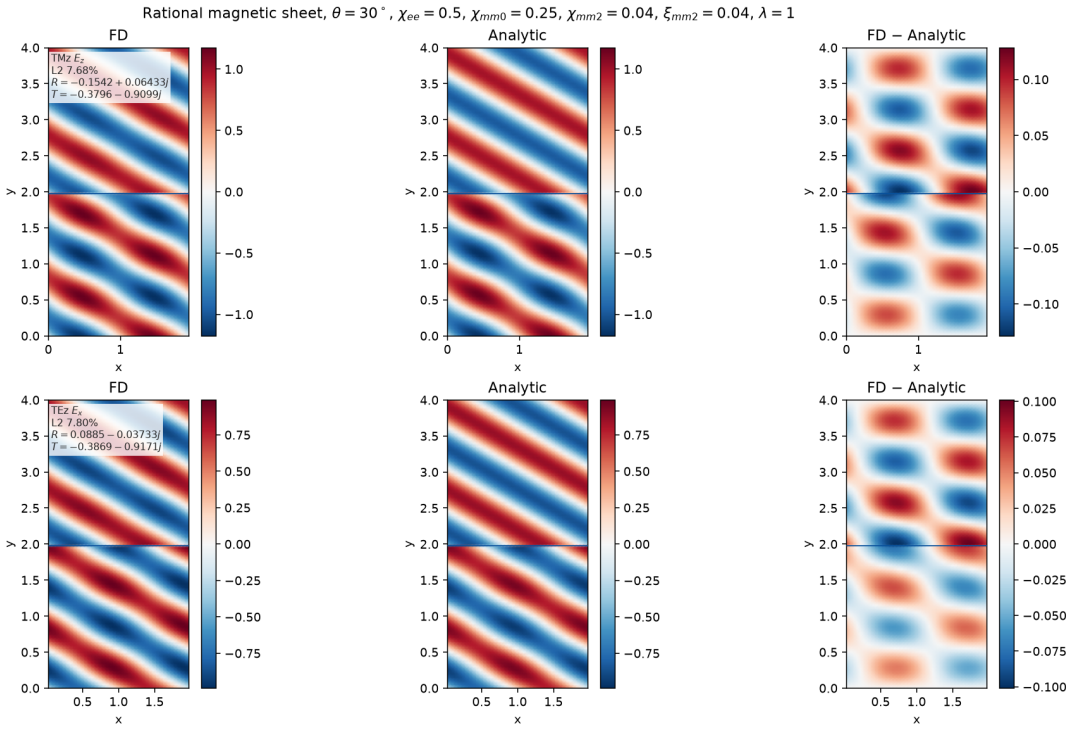

Figure 45: Rational magnetic sheet at $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, $\chi_{mm2} = 0.04$, $\xi_{mm2} = 0.04$, $\lambda = 1$, $\theta = 30°$, and a height of 4λ, on a cell one transverse period wide at $\Delta = 0.05$. The rows are $\mathrm{Re}\, E_z$ (TMz) and $\mathrm{Re}\, E_x$ (TEz). The columns are the Yee field, the analytic field, and their difference. Both polarizations use $\chi_{mm}(k \sin\theta)$. The annotated L^2 is the packed free electric vector.

6 Tests and later slices

The suite is `pytest` from the project root, one hundred seven tests in `tests/`. Milestone 1 checks the curls, the adjoint, the cavity eigenvalues, the manufactured solve, the invariant and the CFL growth, and the fully discrete dispersion, with relative or absolute bars from 10^{-12} to 10^{-6} as listed in the formulation note. Milestone 2 adds the manufactured solve with $\sigma > 0$ and with a smooth $\varepsilon(x, y)$, a uniform $\varepsilon_r = 4$ cavity, the slab-loaded cavity, the lossless staircase invariant, Fresnel face continuity, the Mie continuity and PEC coefficient, the Fresnel and cylinder L^2 bars of 2×10^{-2}, and one period of Debye and Lorentz evolution. The PML tests check the grading, the stretched curls, a zero-thickness step, the line current, and the leapfrog phasor. The formulation note records those bars: a coarse interior error below 4×10^{-2}, a $2.5\times$ drop when the spacing is halved on the same physical band, a bare PEC window above 0.5, and a phasor error below 1.5×10^{-2} at 48 steps per period. The conductor tests compare a PEC band with the shorter cavity it leaves, a PMC quarter-wave with the sine formula, and an open PEC cylinder with the Mie series. The eigenvalue bars are 10^{-10}. The open PEC bars are 8×10^{-2} (TMz) and 10^{-1} (TEz), on the 64^2 grid of Sec. 5.4. The open dielectric bars, at $n = 1.5$ on that same grid, are 8×10^{-3} (TMz) and 1.3×10^{-2} (TEz). That comparison keeps the unstretched samples at least 2Δ off the rim, inside and outside the disk. The 3D curl bars are 10^{-12}. On an $8 \times 6 \times 5$ PEC box the three pure modes match the semi-discrete frequency, with the z term included, to 10^{-10}. A manufactured solve on a $6 \times 5 \times 4$ box recovers the electric vector to the same bar. The stretched 3D curls match the array form to 10^{-12} on that box. The 3D Mie series is continuous across the sphere surface to 10^{-8}. The open dielectric sphere, at $n = 1.5$ on an 18^3 box with a PML of 3 cells, stays under 1.8×10^{-2}. The Hertzian dipole, with $I\ell = 1$ on a 16^3 box at $\Delta = 0.1$, $\lambda = 1$, and a PML of 3 cells, stays under 0.16 outside the band and at least 0.25λ from the element. The Gaussian beam, with $w_0 = 1.5$ on an 8×6 box at $\Delta = 0.05$, $\lambda = 1$, and a PML of 8 cells, stays under 0.10 on the total-field side, outside the band and at least 0.5λ from the split at $y = 1.5$. The scattered-field side of that window stays under 2×10^{-3}. The line-current far field, on a 3λ square at $\Delta = 0.05$, $\lambda = 1$, a PML of 8 cells, and a contour of half-width 0.8λ, stays under 0.05 at $\rho = 20\lambda$. The same physical contour at $\Delta = 0.025$, with a PML of 16 cells, stays under 0.02 and drops the error by more than half. The resistive sheet, at $Z_s = \eta_0$ on an 8×80 box with $\Delta = 1$, $\lambda = 46$, and the sheet on the electric node $y = 40$, stays under 6×10^{-4} (TMz, E_z) and 1.2×10^{-2} (TEz, E_x and E_y). The symmetric sheet, on that box with the face $y = 40.5$, $\chi_{ee} = 0.5$, and $\chi_{mm} = 0.25$, stays under 4.2×10^{-2} (TMz) and 1.9×10^{-1} (TEz). A cell of $\Delta = 1/3$ on the same face drops both errors by more than half. A finite sheet whose samples stay outside the PML is accepted, and a sheet that enters the band is refused. On a 4λ absorber at 10 cells per wavelength, with $\chi_{ee} = \chi_{mm} = -2j/k$ and $\theta = 0$, the lit-side mean $|E|$ stays above 0.95 and the shadow mean stays under 0.20, in both polarizations. The 10λ figure is the example in Sec. 5.13. The normal susceptibility, on a periodic cell one transverse period wide at 30°, with $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi^{nn}_{mm} = 0.4$, and a height of 4λ, stays under 0.24 (TMz) and 0.20 (TEz) at 10 cells per wavelength. Four times that resolution drops both errors by more than half. With the tangential susceptibilities at zero and $\chi^{nn}_{mm} = 0.4$, the TMz error at 20 cells per wavelength stays under 0.17. The example in Sec. 5.14 annotates 9.63% and 8.91% at 20 cells per wavelength. The 10λ sheet with those susceptibilities at 30° is the open example in Sec. 5.16. Its lit and shadow means stay with that figure. The magneto-electric pair, on that same periodic cell with $\chi_{em} = \chi_{me} = 0.2$ added to $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, and $\chi^{nn}_{mm} = 0.4$, stays under 0.24 (TMz) and 0.21 (TEz) at 10 cells per wavelength. Four times that resolution drops both errors by more than half. With only $\chi_{em} = \chi_{me} = 0.2$, the error at 20 cells per wavelength stays under 0.16 in both polarizations. The example in Sec. 5.17 annotates 8.38% and 7.62% at 20 cells per wavelength. The 10λ sheet with those susceptibilities at 30° is the open example in Sec. 5.18. The Bloch cell, 0.8λ wide and 4λ tall at 30°, with $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$, $\chi^{nn}_{mm} = 0.4$, and `Boundary.PERIODIC_X`, stays under 0.22 (TMz) and 0.18 (TEz) at 10 cells per wavelength.

Four times that resolution drops both errors by more than half. The same cell with phase one stays above 0.5 in both polarizations. The Floquet orders $m = \pm 1$, on the electric row a quarter of the way up the cell, stay under 10^{-8} of the specular amplitude. The example in Sec. 5.15 annotates 9.63% and 8.91% at 20 cells per wavelength. A piecewise sheet, 6λ long at 10 cells per wavelength and split into $3\lambda + 3\lambda$, with a PML of 6 cells, compares each piece with that piece's analytic field. The windows sit one wavelength inside the ends and the junction. The left piece is the absorber $\chi_{ee} = \chi_{mm} = -2j/k$ and the right piece is $\chi_{ee} = 0.5$, $\chi_{mm} = 0.25$. In both polarizations the lit-side relative L^2 stays under 0.25 on the absorber and under 0.18 on the right piece. The absorber shadow is an rms of $|E|$, because that transmission is zero, and it stays under 0.20. The right-hand shadow stays under 0.55. That shadow matches a uniform sheet of the same susceptibilities: the ends diffract. The 10λ figure is the example in Sec. 5.19. Polynomial electric dispersion, with $\chi_{ee0} = 0.5$, $\chi_{ee2} = 0.01$, and $\chi_{mm} = 0.25$ on a periodic cell one transverse period wide at 30° and a height of 4λ, stays under 0.22 (TMz) and 0.18 (TEz) at 10 cells per wavelength. TMz is compared with $\chi_{ee0} + \chi_{ee2} k^2 \sin^2\theta$. TEz is compared with χ_{ee0}. Four times that resolution drops both errors by more than half. A sheet that drops χ_{ee2} misses the TMz target by more than the dispersive operator does, at both resolutions. The example in Sec. 5.20 annotates 9.62% and 8.91% at 20 cells per wavelength. Polynomial magnetic dispersion, with $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, and $\chi_{mm2} = 0.01$ on that same periodic cell, stays under 0.18 in both polarizations at 10 cells per wavelength. Both are compared with $\chi_{mm0} + \chi_{mm2} k^2 \sin^2\theta$. Four times that resolution drops both errors by more than half. A sheet that drops χ_{mm2} misses that target by more than the operator does, at both resolutions and in both polarizations. The example in Sec. 5.21 annotates 8.35% and 8.12% at 20 cells per wavelength. The magnetic denominator, with $\chi_{ee} = 0.5$, $\chi_{mm0} = 0.25$, $\chi_{mm2} = 0.04$, and $\xi_{mm2} = 0.04$ on that same periodic cell, stays under 0.18 (TMz) and 0.21 (TEz) at 10 cells per wavelength. Both are compared with $(\chi_{mm0} + \chi_{mm2} k^2 \sin^2\theta)/(1 + \xi_{mm2} k^2 \sin^2\theta)$. Four times that resolution drops both errors by more than half. A polynomial sheet, $\xi_{mm2} = 0$, misses that target by more than the operator does, at both resolutions and in both polarizations. The example in Sec. 5.22 annotates 7.68% and 7.80% at 20 cells per wavelength.

A failing bar is a reason to refine the grid in that test, not to widen the bar.

The formulation note already names the sheet work that remains to be coded: Lorentz poles on the sheet. The Fresnel and dielectric-cylinder domain figures still draw an orange thickness as a layout. Those two solves use the Dirichlet trace. The open PEC cylinder of Sec. 5.4, the open dielectric cylinder of Sec. 5.5, the open dielectric sphere of Sec. 5.7, the Hertzian dipole of Sec. 5.8, the Gaussian beam of Sec. 5.9, the line-current far field of Sec. 5.10, the finite absorbing sheet of Sec. 5.13, the finite oblique sheet of Sec. 5.16, and the finite magneto-electric sheet of Sec. 5.18 are the solves that stretch the derivatives.

References

[1] K. S. Yee, "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media," *IEEE Trans. Antennas Propag.*, 1966.

[2] T. J. Smy, S. Stewart, and S. Gupta, "Integrated generalized sheet transition conditions in a Yee-cell based FDTD simulation of electromagnetic metasurfaces," arXiv:1706.10136, 2017.